

Introduction

1 min

Imagine that a California state law was passed in 2016 that raised the minimum wage beginning in the year 2017. We are interested in what this law's impact has been on student wages since student jobs often pay low wages.

We have the data on the average student wages at all California public universities from the year 2007 to 2017, displayed in the plot in the learning environment. We note that wages rise and fall from year to year, but we see a particularly large rise from 2016 to 2017.

It's easy to assume the large increase was entirely due to the new minimum wage law. But what would average student wages in 2017 have looked like had the law NOT been passed?

- Would wages still have increased a lot, meaning the law had little impact?
- Would wages have decreased, meaning the law had an even larger impact than just the difference between 2016 and 2017 wages?

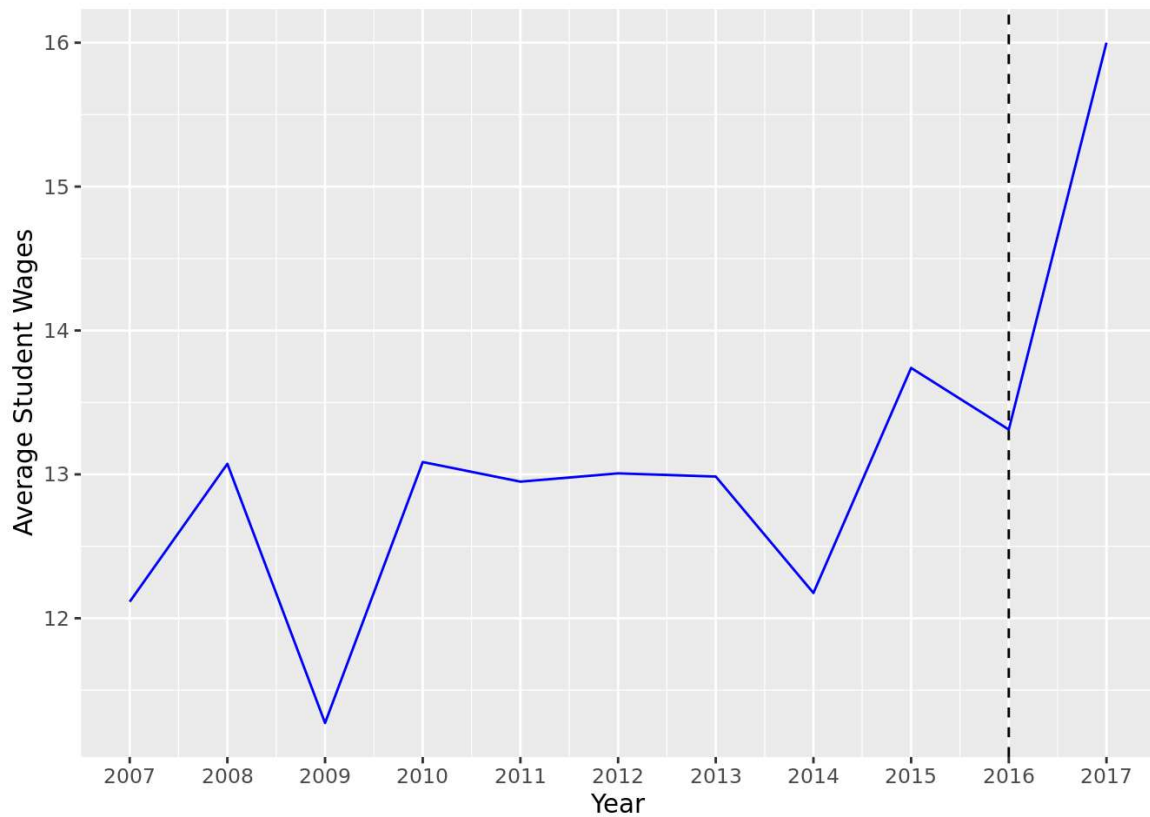
The fundamental principle of causal inference tells us that we can't observe both situations, so we need to use other observed data as a substitute for what student wages in California would have looked like in 2017 had the law not passed.

Difference in differences (DID) is a causal inference technique that estimates a treatment effect by analyzing treatment and control trends over time. In the following exercises, we will walk through how to apply DID to our student wage example, review the different assumptions and limitations of the DID

[method](#)

Preview: Docs A method is a small piece of code, usually defined in a class, that can be used outside the class and in other parts of the program.

, and learn how to perform this method using R programming.



Working Example of DID

12 min

Let's say we have some data on student wages at the state level in a dataset called `wages`. It has the following variables:

- `state`: state where the universities are located
- `year`: year the data is from
- `avg_wage`: average student wage for all public universities in the state

Start by using the package `ggplot2` in R to re-create the student wage line plot from the last exercise. Since we only want to plot California schools, filter by the `state` variable and then add code for our plot.

```
# import libraries
library(dplyr)
library(ggplot2)
# plot wages versus years
ca_wages <- wages %>%
  #only California schools
  filter(state == "California") %>%
  #wages over time
  ggplot(
    aes(x = year, y = avg_wage)) +
  #line plot
```

```
geom_line()
```

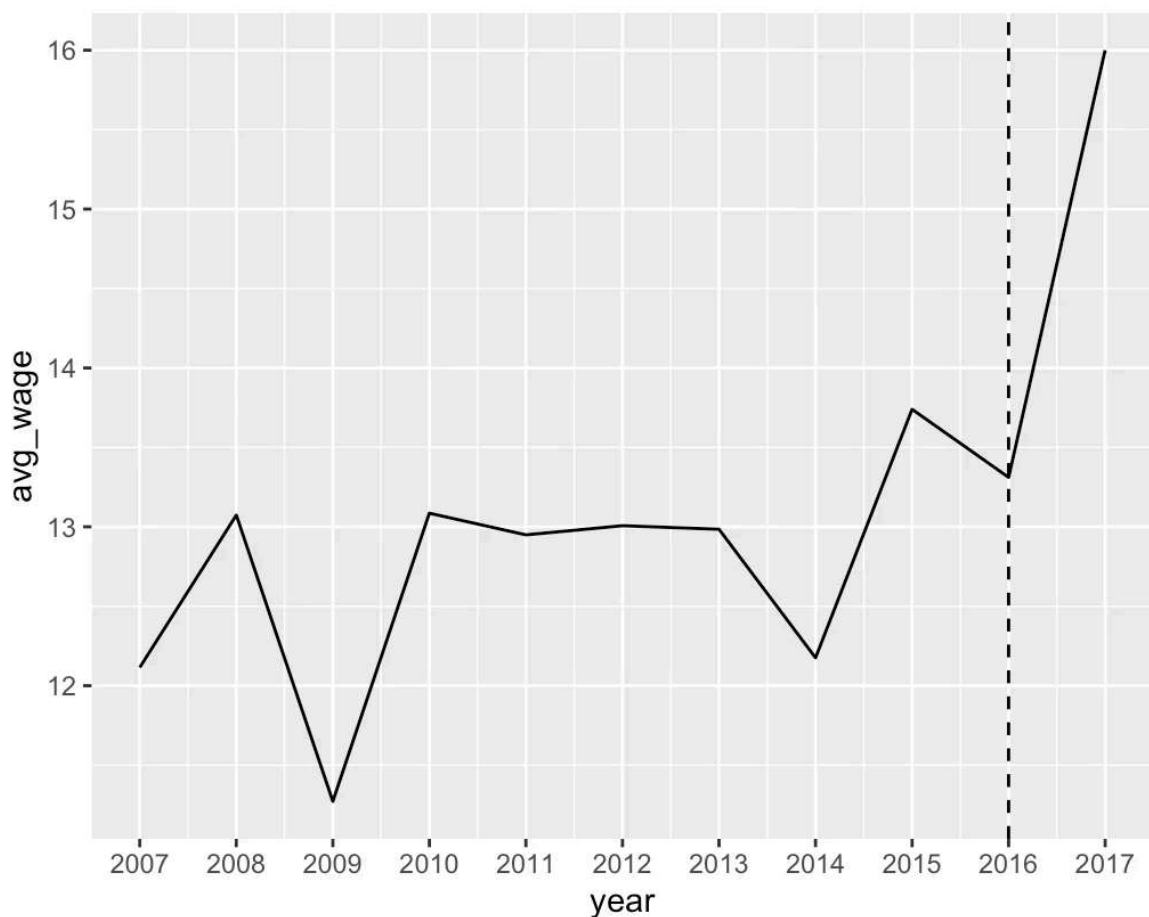
Copy to Clipboard

The situation created by the law is a *natural experiment*. Rather than having a researcher randomly assign treatment and control groups to study minimum wage effects, treatment assignment is decided by some outside force. In this case, that outside force was the minimum wage law that went into effect in 2017.

Let's add a dashed vertical line to our plot to separate the time before and after the law went into effect. We'll also label the x-axis scale to see the years more clearly.

```
ca_wages +  
  geom_vline(xintercept = 2016, linetype = "dashed") +  
  scale_x_continuous(breaks = c(2007:2017))
```

Copy to Clipboard



We can see that there is some change after the law is implemented in 2016, but we don't know if the change is due to the law or because of other conditions that happened at the same time. We need some data to use as a counterfactual to this situation: what student wages in California would have looked like if the law never happened.

Instructions

1. Checkpoint 1 Passed

1.

Let's say there was a new entertainment tax in Sydney starting in 2019. You want to find out if the tax affected movie theater ticket sales. You have data about average annual movie theater ticket sales in Sydney from 2012 through 2019 with the following variables:

This data is contained in the dataset `tickets` which has been loaded for you in **notebook.Rmd** with the first few rows printed for you in the workspace.

Make a line plot that shows average movie tickets for Sydney by year. Remember to filter

`city` to look at just Sydney. Save the plot as `syd_sales`.

- `city` : city where the universities are located
- `year` : year the data is from
- `sales` : average ticket sales for theaters in the city

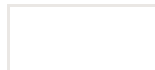
2. Checkpoint 2 Passed

2.

Add to `syd_sales` a dashed vertical line at x=2018 and x-axis scale labels for the years 2012 to 2019. What happened to ticket sales in the year the tax was implemented?

Working Example of DID

```
# import libraries
library(dplyr)
library(ggplot2)
# import dataset
tickets <- read.csv("ticket_sales.csv")
# view dataset
head(tickets)
```



	city <fct>	year <int>	sales <dbl>
1	Sydney	2012	516.1746
2	Sydney	2013	517.2626
3	Sydney	2014	514.8101
4	Sydney	2015	515.9932
5	Sydney	2016	513.2907
6	Sydney	2017	515.4289

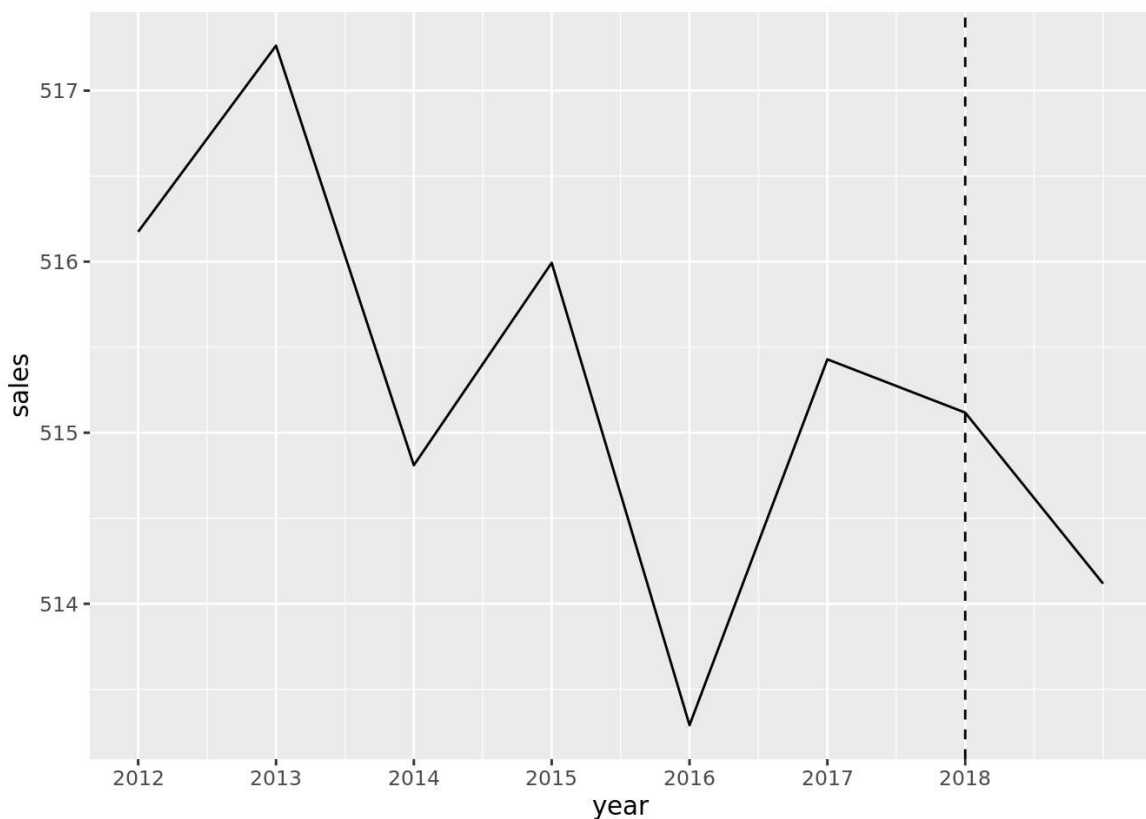
6 rows

Checkpoint 1

```
# create plot of ticket sales
syd_sales <- tickets %>%
  #only Sydney
  filter(city == "Sydney") %>%
  #sales over time
  ggplot(aes(x = year, y = sales)) +
  #line plot
  geom_line()
```

Checkpoint 2

```
# add line at x=2018 and x-axis label
syd_sales +
  geom_vline(xintercept = 2018, linetype = "dashed") +
  scale_x_continuous(breaks = c(2009:2018))
```



title: "Working Example of DID"

output: html-notebook

```
```{r, message=FALSE}

import libraries

library(dplyr)

library(ggplot2)

import dataset

tickets <- read.csv("ticket_sales.csv")

view dataset

head(tickets)

...

Checkpoint 1

```{r}

# create plot of ticket sales

syd_sales <- tickets %>%

  #only Sydney

  filter(city == "Sydney") %>%

  #sales over time

  ggplot(aes(x = year, y = sales)) +

  #line plot

  geom_line()

...

# Checkpoint 2

```{r}

add line at x=2018 and x-axis label

syd_sales +

 geom_vline(xintercept = 2018, linetype = "dashed") +

 scale_x_continuous(breaks = c(2012:2019))

...


```

## Finding a Control Group

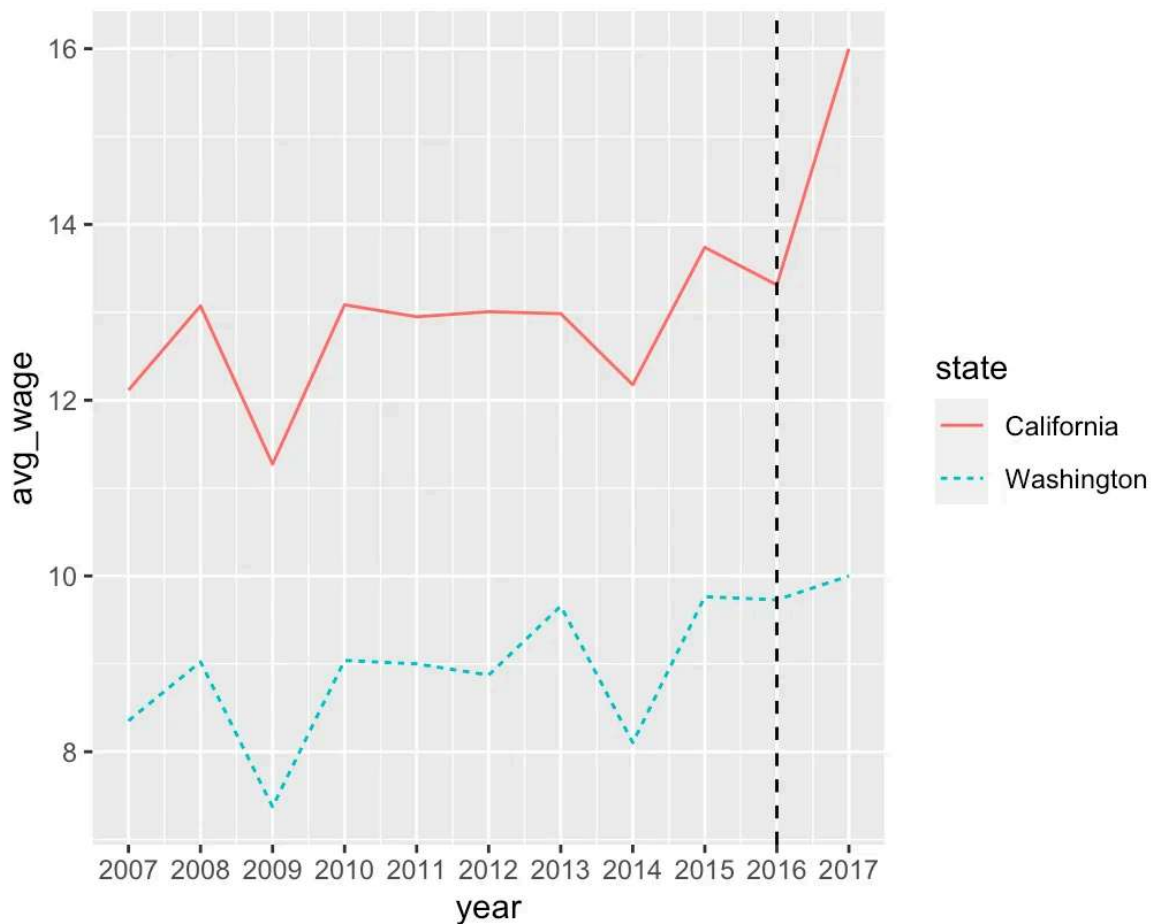
8 min

It is impossible to know what truly would have happened in California schools if the law had never happened. However, we can look for another state whose average student wages that follow a similar pattern over time — but did not receive the same treatment. We can use the other state's data to approximate the counterfactual. With DID, we use the *trend* of a control group to estimate the counterfactual trend for the treated. In other words, we use DID to estimate the average treatment effect on the treated (ATT).

In this example, a control trend might be the average student wage of universities in Washington, given that Washington did not implement a new minimum wage law. We can view these two groups in a single plot using the following code:

```
import library
library(ggplot2)
plot wages versus years by state
ggplot(data = wages, aes(x = year, y = avg_wage, color = state, linetype
= state)) +
 geom_line() +
 #vertical line at 2016
 geom_vline(xintercept = 2016, linetype = "dashed") +
 #x-axis scale 2007 to 2017
 scale_x_continuous(breaks = c(2007:2017))
```

Copy to Clipboard



We can see that in the years leading up to 2016, the average minimum wage at Washington colleges followed a very similar pattern to the California colleges, only shifted lower. We hope that these similar patterns account for other important

[variables](#)

Preview: Docs Loading link description

, such as the impact on wages of living costs and necessary student expenses.

## Instructions

### 1. Checkpoint 1 Passed

1.

The `tickets` dataset also contains data on average ticket sales in Toronto. Create a line plot that shows the ticket sales for both Toronto and Sydney over the same years. Be sure to change the color and type of line by city. Does Toronto have a similar trend in ticket sales from 2012 to 2018?

Note: you can optionally add a dashed vertical line at 2018 and set the x-axis scale using the functions in the last two lines of code shown in the narrative.

---

title: "Compare Treatment and Control"

output: html-notebook

---

```
```{r, message=FALSE}

# import library

library(ggplot2)

# import dataset

tickets <- read.csv("ticket_sales.csv")

...

# Checkpoint 1

```{r}

plot ticket sales for Sydney and Toronto

ggplot(data = tickets, aes(year, sales, color = city, linetype = city)) +

 geom_line() +

 #vertical line at 2018

 geom_vline(xintercept = 2018, linetype = "dashed") +

 #x-axis scale 2007 to 2017
```

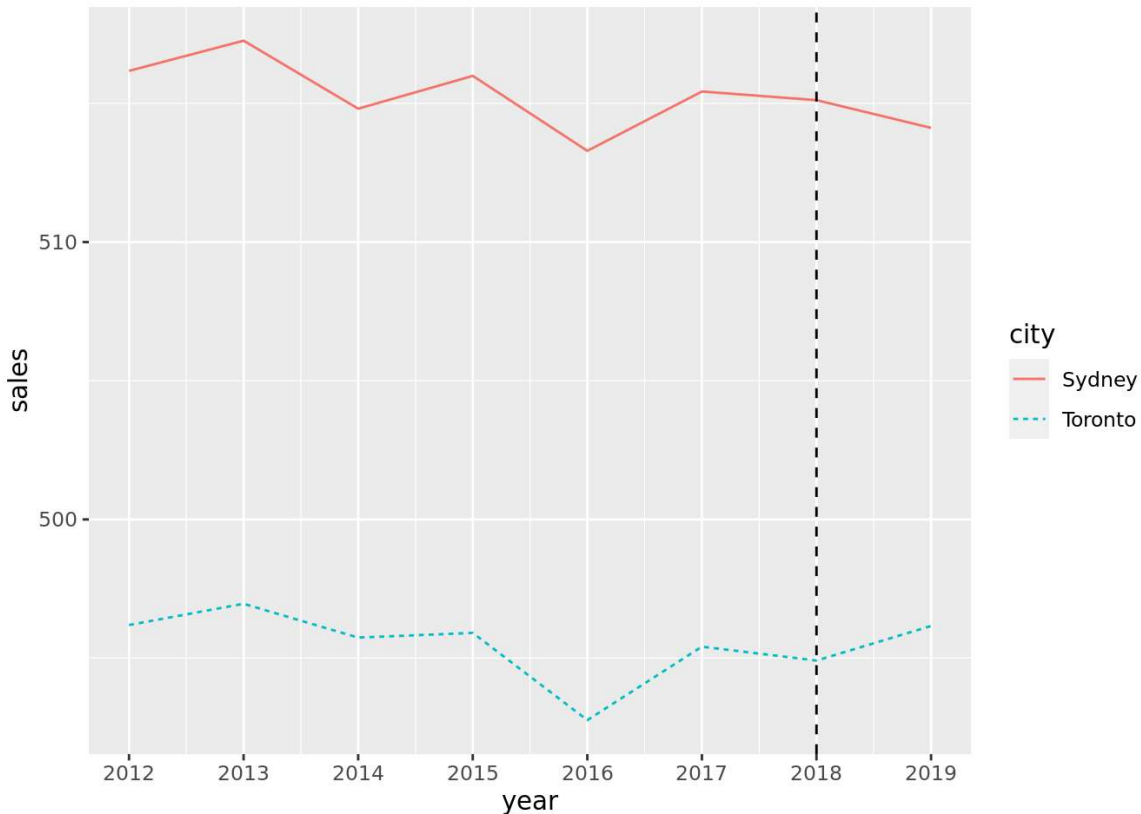
```
scale_x_continuous(breaks = c(2012:2019))
```

```
...
```

# Compare Treatment and Control

## Checkpoint 1

```
plot ticket sales for Sydney and Toronto
ggplot(data = tickets, aes(year, sales, color = city, linetype = city)) +
 geom_line() +
 #vertical line at 2018
 geom_vline(xintercept = 2018, linetype = "dashed") +
 #x-axis scale 2007 to 2017
 scale_x_continuous(breaks = c(2012:2019))
```



## Assumptions

2 min

We noted that Toronto may be a good control group for Sydney because the trends in ticket sales follow a similar pattern. This is actually a necessity of DID called the *parallel trends* assumption: trends between the control and treatment groups should be similar prior to the start of treatment.

The specific values between groups do not need to be the same, but the two groups should have the same change over time. We can check this visually by plotting the data by group and verifying that the general trends are similar, as we saw in the previous exercise. A disadvantage to DID is the difficulty of meeting the parallel trends assumption: we need pre-treatment data for both groups AND the data has to show similar trends before treatment.

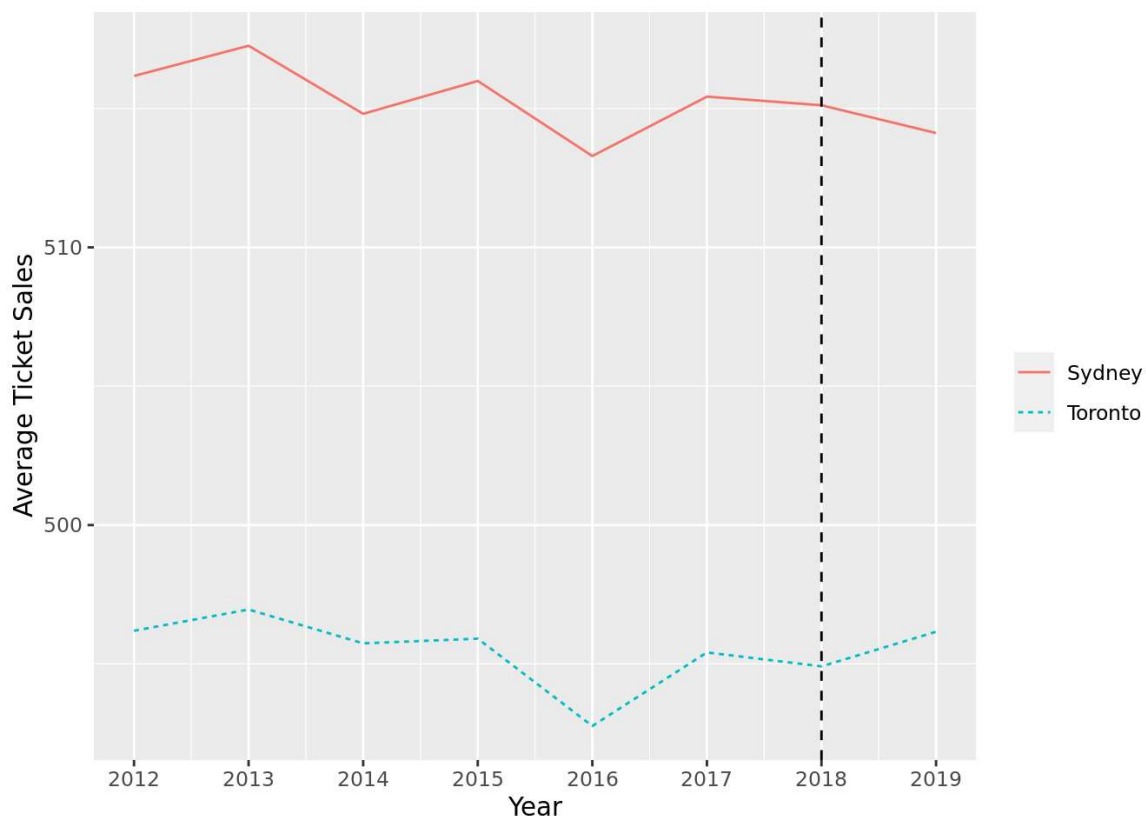
Another assumption for DID is that the treatment group was selected for the treatment by a factor that is independent of the outcome. This assumption is met for our example: the treatment is determined by state, which is independent of the overall average student wages.

Finally, the assumptions for DID also include the normal assumptions for all causal inference methods:

- Conditional exchangeability
- Stable Unit Treatment Value Assumption (SUTVA)
- Overlap

#### Instructions

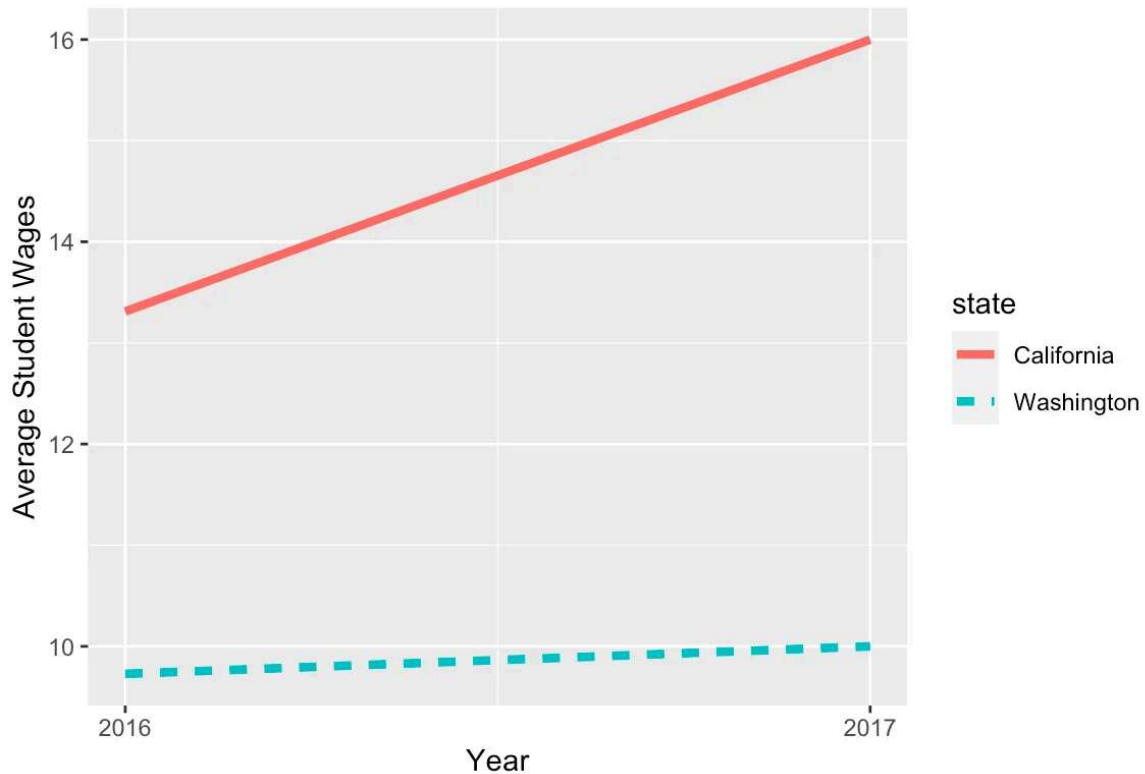
1. The learning environment shows your plot from the ticket sales data in the previous exercise. Take a look at the lines for Toronto and Sydney covering the time from 2012 to 2018, just before the new tax was implemented. Both lines follow a similar trend going up and down slightly each year but with a particular dip in 2016.
2. Do you think Toronto will make a good approximation of Sydney's counterfactual even though its line is lower on the plot?
3. Do you think we've met the other assumptions of DID in this case?



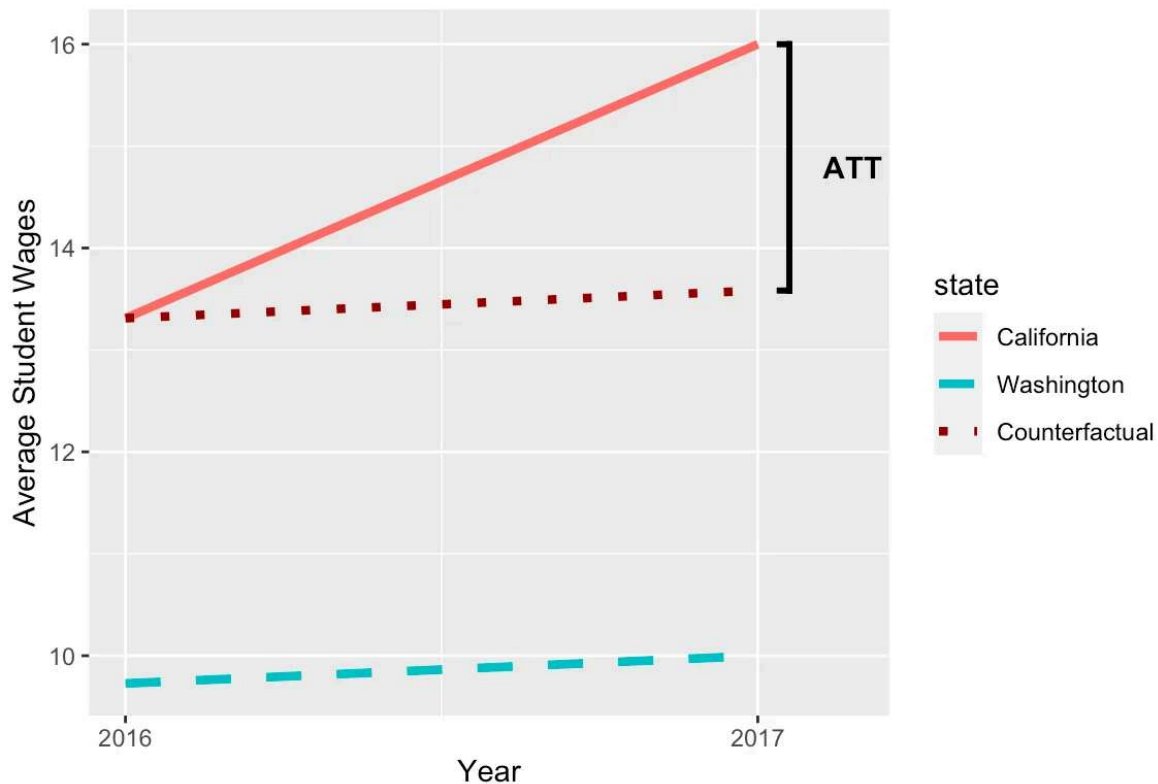
## The 2x2 Case

3 min

Normally we would want to use all the data we have available to us to perform our DID model. However, that type of analysis can be complex. To keep things simple, we are going to focus on a 2x2 analysis for our student wage data, which means we will look at differences in average wages from the year before the law went into effect (2016) and the year after (2017).



From this plot, we see that the Washington schools had a much smaller increase in average student wages from 2016 to 2017 than the California schools did. We assume that if the law had NOT been passed, the pattern of student wages for the California schools would have been similar to the pattern of wages for the Washington schools. In other words, we assume the slope of the California wages would be the same as that of the Washington wages, as illustrated by the dotted red line in the following plot.



With this new line based on Washington's pattern, we have drawn our approximation of California's counterfactual trend. The difference between California's 2017 observed wages and its 2017 counterfactual wages will be our estimated ATT, as shown in the plot. In the next exercises, we will compute this treatment effect using the approximated counterfactual trend in two ways.

#### Instructions

##### 1. Checkpoint 1 Passed

1.

A new dataset called `tickets2` that only includes the data from 2018 and 2019 has been created for you in `notebook.Rmd`. Uncomment and run the code in the workspace to view a plot of average tickets sales for Sydney and Toronto in those two years.

Based on the plot, consider the following:

- Do you think average ticket sales for Sydney would have increased or decreased had their not been a new tax?
- What do you think the impact of the tax was on average ticket sales in Sydney?

---

title: "The 2x2 Case"

output: html-notebook

---

```
``{r, message=FALSE}
```

```
import libraries
```



```
library(ggplot2)

library(dplyr)

import dataset

tickets <- read.csv("ticket_sales.csv")

subset to 2018 and 2019

tickets2 <- tickets %>%

 filter(year > 2017)

...

Checkpoint 1

```{r}

# Uncomment to view plot

ggplot(data = tickets2, aes(x = year, y = sales, color = city, linetype = city)) +

  geom_line(size = 1.5) +

  scale_x_continuous(breaks = c(2018, 2019))

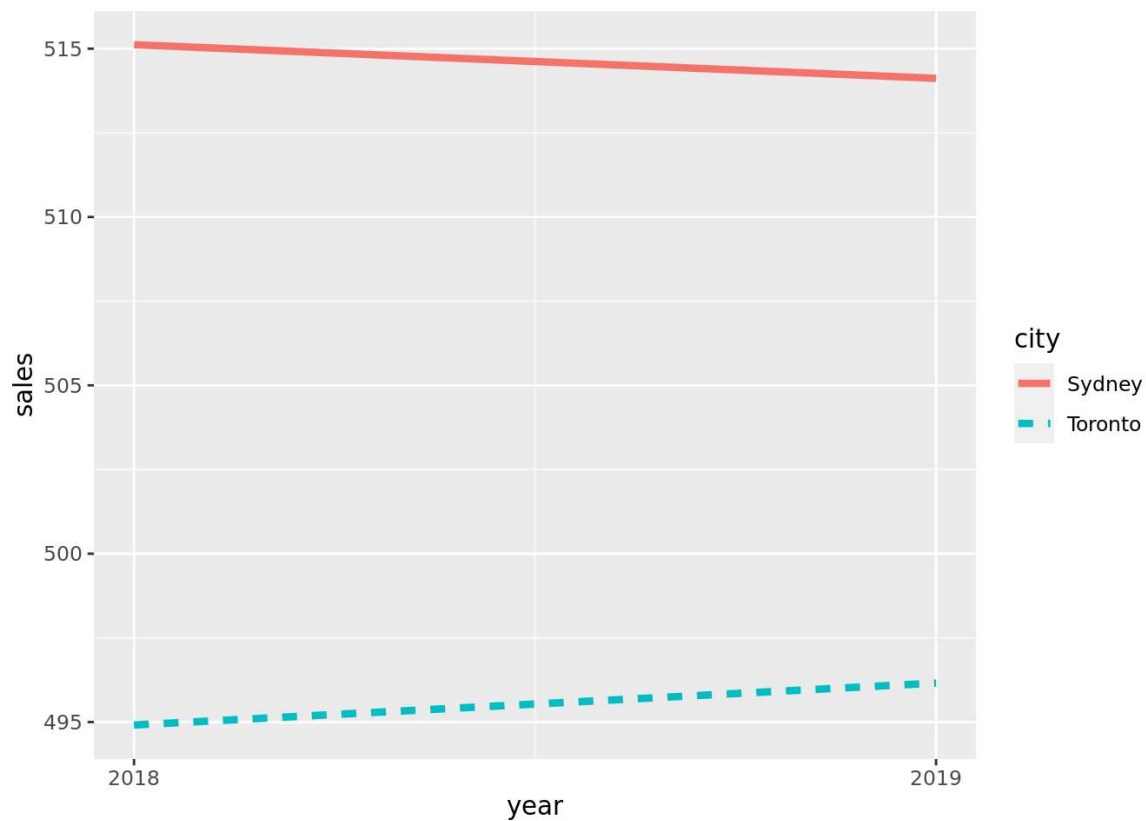
...
```

The 2x2 Case

```
# import libraries
library(ggplot2)
library(dplyr)
# import dataset
tickets <- read.csv("ticket_sales.csv")
# subset to 2018 and 2019
tickets2 <- tickets %>%
  filter(year > 2017)
```

Checkpoint 1

```
# Uncomment to view plot
ggplot(data = tickets2, aes(x = year, y = sales, color = city, linetype =
city)) +
  geom_line(size = 1.5) +
  scale_x_continuous(breaks = c(2018, 2019))
```



Estimate Using Difference in Means

8 min

One way we can estimate the ATT is by comparing the differences in average wages over both time and treatment groups.

Let's view the mean wages for both states for 2016 and 2017 only.

```
# import library
library(dplyr)
# filter data
wages %>%
  filter(year > 2015) #only 2016 and 2017
```

Copy to Clipboard

The output looks like the following:

```
state year avg_wage
1 California 2016 13.311279
2 California 2017 16.000000
3 Washington 2016 9.728146
4 Washington 2017 10.000000
```

```
Copy to Cli
```

```
pboard
```

First, let's use the means in our output to take the difference in average wages before and after the law for each state.

California

$$\text{mean}_{2017} - \text{mean}_{2016} = \$16.00 - \$13.31 = \$2.69$$

Washington

$$\text{mean}_{2017} - \text{mean}_{2016} = \$10.00 - \$9.73 = \$0.27$$

Hypothetically, if the law had no effect on wages, we would expect the difference between the post-treatment mean and the pre-treatment mean for the treatment group would be similar to the difference for the control group. However, these findings match what we saw in our plots before: California experienced a MUCH LARGER change in average wages from 2016 to 2017 than Washington did.

But how much of that California change was due to factors other than the new law? We answer this question by taking the difference of the differences we just found — which is where this method gets its name!

$$\text{ATT} = \text{difference}_{\text{California}} - \text{difference}_{\text{Washington}}$$

$$\text{ATT} = \$2.69 - \$0.27$$

$$\text{ATT} = \$2.42$$

According to our basic 2x2 DID that uses mean differences alone, we estimate the average treatment effect on the treated to be \$2.42. But there's still another DID method we might want to try!

Instructions

1. Checkpoint 1 Passed

The `tickets` dataset has been filtered to just the years 2018 and 2019 and saved for you as `tickets2` in `notebook.Rmd`. Print `tickets2` and inspect the average ticket sales for each city.

2. Checkpoint 2 Passed

Using the `tickets2` output and either R or a calculator, compute the change in average tickets sales from 2018 to 2019 for each city. Save Toronto's difference as `Tdiff` and Sydney's difference as `Sdiff`.

3. Checkpoint 3 Passed

Compute the DID estimate for the impact of the new tax by subtracting Toronto's difference from Sydney's difference. Save this value as `did_means`.

4. Checkpoint 4 Passed

Print `did_means` and consider its value. What does this tell you about the impact of the new tax on average ticket sales in Sydney?

title: "Estimate Using Difference in Means"

output: html-notebook

```
```{r, message=FALSE}
```

```
import libraries
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
import dataset
```

```
tickets <- read.csv("ticket_sales.csv")
```

```
subset to 2018 and 2019
```

```
tickets2 <- tickets %>%
```

```
 filter(year > 2017)
```

```
...
```

```
Checkpoint 1
```

```
```{r}
```

```
# print tickets2
```

```
print(tickets2)
```

```
...
```

```
# Checkpoint 2
```

```
```{r}
```

```
save values of Tdiff and Sdiff
```

```
Tdiff <- tickets2$sales[4] - tickets2$sales[3]
```

```
Sdiff <- tickets2$sales[2] - tickets2$sales[1]
```

```
rounded alternative:
```

```
Tdiff <- 496.16 - 494.91
```

```
Sdiff <- 514.12 - 515.12
```

```
...
```

```
Checkpoint 3
```

```
```{r}
```

```
# subtract Tdiff from Sdiff
```

```
did_means <- Sdiff - Tdiff
```

```
...
```

```
# Checkpoint 4
```

```
```{r}
```

```
print did_means
```

```
print(did_means)
```

```
...
```

## Estimate Using Difference in Means

```
import libraries
library(ggplot2)
library(dplyr)
import dataset
tickets <- read.csv("ticket_sales.csv")
subset to 2018 and 2019
tickets2 <- tickets %>%
 filter(year > 2017)
```

## Checkpoint 1

```
print tickets2
print(tickets2)
```

```
 city year sales
1 Sydney 2018 515.1175
2 Sydney 2019 514.1180
3 Toronto 2018 494.9121
4 Toronto 2019 496.1579
```

## Checkpoint 2

```
save values of Tdiff and Sdiff
Tdiff <- tickets2$sales[4] - tickets2$sales[3]
Sdiff <- tickets2$sales[2] - tickets2$sales[1]
rounded alternative:
```

```
Tdiff <- 496.16 - 494.91
Sdiff <- 514.12 - 515.12
```

## Checkpoint 3

```
subtract Tdiff from Sdiff
did_means <- Sdiff - Tdiff
```

## Checkpoint 4

```
print did_means
print(did_means)
```

```
[1] -2.245214
```

## Estimate Using Linear Regression

11 min

While we were able to estimate the ATT through mean differences alone, we can also use linear regression for DID. A simple DID regression model predicts the outcome from the

[variables](#)

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for treatment group and time, along with the interaction of treatment with time.

To simplify our output, we will first transform the `state` variable to a treatment indicator called `treat` (1 for California and 0 for Washington). Then we transform `year` to a time indicator called `time` (1 for 2017 and 0 for 2016).

```
transform state to treat
wages2$treat <- ifelse(wages2$state=="California",1,0)
transform year to time
wages2$time <- ifelse(wages2$year==2017,1,0)
```

Copy to Clipboard

We can print and inspect our dataset to make sure our transformations were done correctly.

```
 state treat year time avg_wage
1 California 1 2016 0 13.311279
2 California 1 2017 1 16.000000
```

```
3 Washington 0 2016 0 9.728146
4 Washington 0 2017 1 10.000000
```

Copy to Clipboard

To create a DID regression for our student wage data, we run a model that predicts average student wages from the treatment, time, and the interaction of treatment and time. Note that in R, `treat*time` is equivalent to `treat + time + treat:time`.

```
did_mod <- lm(
 #include interaction
 avg_wage ~ treat*year,
 #use subsetting data
 data = wages2
)
```

Copy to Clipboard

When we print `did_mod` we get the following output. Note that the coefficient on the interaction term `treat:year` is exactly what we computed for the ATT by taking the difference of means. We estimate the impact of the minimum wage law on California student wages to be an increase of about \$2.42.

```
Call:
lm(formula = avg_wage ~ treat * time, data = wages2)

Coefficients:
(Intercept) treat time treat:time
 9.7281 3.5831 0.2719 2.4169
```

Copy to Clipboard

What do all the other coefficients represent?

- **Intercept** indicates the expected value for pre-treatment average student wages for the control group (Washington 2016).
- **treat** is the difference between the control group and the treatment group at the pre-treatment time (California 2016 - Washington 2016).
- **time** is the difference between the pre-treatment and post-treatment times for the control group (Washington 2017 - Washington 2016).

Combinations of these coefficients give us back all four means from our dataset.

Mean	Equivalent Coefficient(s)
Washington 2016	Intercept

California 2016	Intercept + treat
Washington 2017	Intercept + time
California 2017	Intercept + treat + time + treat:time

### Instructions

#### 1. Checkpoint 1 Passed

1.

The dataset `tickets2` with only 2018 and 2019 data has been saved for you in `notebook.Rmd`. Use the `ifelse()` function to add to `tickets2` the indicator variable `treat` that is 1 for Sydney and 0 for Toronto.

#### 2. Checkpoint 2 Passed

2.

Use the `ifelse()` function to add to `tickets2` the indicator variable `time` that is 1 for 2019 and 0 for 2018.

#### 3. Checkpoint 3 Passed

3.

Create a linear regression model `did_reg` that predicts average tickets sales from `treat`, `time`, and the interaction of `treat` with `time`.

#### 4. Checkpoint 4 Passed

4.

Now print `did_reg` and inspect the coefficient on the interaction term. was the direction of the impact what you expected based on your earlier plots?

---

title: "Estimate Using Linear Regression"

output: html-notebook

---

```
```{r, message=FALSE}
```

```
# import library
```

```
library(dplyr)
```

```
# import dataset
```

```
tickets <- read.csv("ticket_sales.csv")
```

```
# filter to only 2018 and 2019
```

```
tickets2 <- tickets %>%
```

```
  filter(year > 2017)
```

```
...
```

```
# Checkpoint 1
```



```

```{r}

add treatment indicator

tickets2$treat <- ifelse(tickets2$city=="Sydney",1,0)

...

Checkpoint 2

```{r}

# add time indicator

tickets2$time <- ifelse(tickets2$year==2019,1,0)

...


# Checkpoint 3

```{r}

save linear model

did_reg <- lm(

 #include interaction

 sales ~ treat*time,

 #subsetting data

 data = tickets2

)

...

Checkpoint 4

```{r}

# view model results

print(did_reg)

...

```

Estimate Using Linear Regression

```

# import library
library(dplyr)
# import dataset

```

```

tickets <- read.csv("ticket_sales.csv")
# filter to only 2018 and 2019
tickets2 <- tickets %>%
  filter(year > 2017)

```

Checkpoint 1

```

# add treatment indicator
tickets2$treat <- ifelse(tickets2$city=="Sydney",1,0)

```

Checkpoint 2

```

# add time indicator
tickets2$time <- ifelse(tickets2$year==2019,1,0)

```

Checkpoint 3

```

# save linear model
did_reg <- lm(
  #include interaction
  sales ~ treat*time,
  #subsetting data
  data = tickets2
)

```

Checkpoint 4

```

# view model results
print(did_reg)

```

Call:

```
lm(formula = sales ~ treat * time, data = tickets2)
```

Coefficients:

(Intercept)	treat	time	treat:time
494.912	20.205	1.246	-2.245

Review

1 min

Congratulations! You've completed a lesson on the fundamentals of difference in differences for causal inference. Through applied exercises you learned:

DID uses the trend of a control group as an approximate counterfactual trend for the treatment group.

DID requires that we have pre-treatment data for both groups and that we meet the parallel trends assumption.

We can compute the ATT by taking mean differences or through linear regression.

You may be wondering why we would use linear regression at all when we can just use means. Regression allows us to control for confounding

variables

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as well as get estimated standard errors. We explored simple models in this lesson to get a strong foundation, but more complex DID analyses benefit from the added flexibility of regression modeling.

Instructions

Check out the output in the learning environment. In this DID analysis, we found that the tax in Sydney reduced average ticket sales by 2.25. What does each coefficient in the model represent for the ticket sales problem?

Estimate Using Linear Regression

	city	year	sales	treat	time
1	Sydney	2018	515.1175	1	0
2	Sydney	2019	514.1180	1	1
3	Toronto	2018	494.9121	0	0
4	Toronto	2019	496.1579	0	1

Call:

```
lm(formula = sales ~ treat * time, data = tickets2)
```

Coefficients:

(Intercept)	treat	time	treat:time
494.912	20.205	1.246	-2.245

What do all the other coefficients represent?

Intercept indicates the expected value for pre-treatment average outcome for the control group.

treat is the difference between the control group and the treatment group at the pre-treatment time.

time is the difference between the pre-treatment and post-treatment times for the control group.

Combinations of these coefficients give us back all four means from our dataset.

Mean

Equivalent Coefficient(s)

Toronto 2018

Intercept

Sydney 2018

Intercept + treat

Toronto 2019

Intercept + time

Sydney 2019

Intercept + treat + time + treat:time

Fill in the code that would create a Difference in Differences model in R for outcome `outcome`, treatment group indicator `treat`, and timepoint variable `time`.

```
did_model <- lm ( outcome ~ treat *time, data = df)
```

 You got it!

Which of the following does the Difference in Differences model estimate?

Average treatment effect on the control (ATC)

Effect of time alone

Average treatment effect on the treated (ATT)

 Yes! We want to know the effect that the treatment has on those who received the treatment.

Which of the following is NOT an assumption for DID?

Conditional exchangeability

The groups within the data need to have parallel trends over time.

Data must be normally distributed.

 The data do not need to be normally distributed for DID to be successful.

Treatment for the treated was not determined by the value of the outcome variable before treatment assignment.

Fill in the spaces to manually calculate the ATT using the Difference in Differences model.

Treatment group change over time:

$$\text{treat_diff} = \text{treatment_post} - \text{treatment_pre}$$

Control group change over time:

$$\text{control_diff} = \text{control_post} - \text{control_pre}$$

Difference in differences:

$$\text{DID} = \text{treat_diff} - \text{control_diff}$$



You got it!

Which of the following coefficients represents the ATT in the Difference in Difference regression output?

(Intercept)	treat	time	treat:time
672.81456	35.831336	27.18544	3.4168664

time

treat

(Intercept)

treat:time



Correct! The ATT estimate is the coefficient on the interaction of treatment group and time.

****Correct Answer:****

*****`treat:time`****

Explanation

In a standard ****Difference-in-Differences (DiD)**** regression model:

```

```r
lm(outcome ~ treat * time, data = df)
```

```

The coefficient on the **interaction term** ``treat:time`` represents the **Average Treatment Effect on the Treated (ATT)**.

Interpretation of Each Coefficient

| Coefficient | Interpretation |
|---------------------|--|
| ----- ----- | |
| (Intercept) | Baseline outcome for the control group in the pre-period |
| `treat` | Pre-treatment difference between treated and control groups |
| `time` | Change over time in the control group (common time trend) |
| `treat:time` | ATT — the <i>additional</i> change in the treated group after treatment (the causal effect of interest) |

Why ``treat:time`` is the ATT

The DiD estimator isolates the causal effect by subtracting the control group's time change from the treatment group's time change:

\$\$

$$\text{ATT} = (\text{Treated}_{\text{post}} - \text{Treated}_{\text{pre}}) - (\text{Control}_{\text{post}} - \text{Control}_{\text{pre}})$$

\$\$

This is exactly what the **``treat:time``** coefficient captures.

Quick Code Reference

```
```r
did_model <- lm(outcome ~ treat * time, data = df)
summary(did_model)
```
```

Look for the row labeled `treat:time` — that is your DiD estimate (ATT).

Which of the following is a disadvantage to DID?

The information required is extensive and highly confusing.

The effects are not easily interpretable.

The model is complex and needs complicated code to implement.

Needing pre-treatment data for both the treatment and control group.



Finding enough pre-treatment data to validate the parallel trend assumption is not always easy.

Correct Answer:

Needing pre-treatment data for both the treatment and control group.

Why This Is a Disadvantage of DiD

Difference-in-Differences (DiD) requires **panel data** (or repeated cross-sections) with observations **before and after** the treatment for **both** the treatment and control groups.

This is a significant limitation because:

- Many real-world settings lack good historical (pre-treatment) data.
- You cannot use DiD for completely new programs without baseline data.
- It increases data collection costs and complexity.

Why the Other Options Are ****Not**** Major Disadvantages

| Statement | Why It's Not a Real Disadvantage |

|-----|-----|

| The information required is extensive and highly confusing | DiD is relatively straightforward conceptually |

| The effects are not easily interpretable | DiD effects are ****very**** interpretable (especially with graphs) |

| The model is complex and needs complicated code | DiD is one of the ****simplest**** models:
`lm(outcome ~ treat \* time)` |

Simple DiD Code Reminder (R)

```
``r
```

```
# Very simple implementation
```

```
did_model <- lm(outcome ~ treat * time, data = df)
```

```
summary(did_model)
```

```
```
```

The key coefficient (`treat:time`) gives you the **\*\*ATT\*\*** — easy to interpret.

---

### ### Trade-offs Summary

**\*\*Advantages of DiD\*\*:**

- Intuitive and easy to explain.
- Controls for time-invariant unobserved heterogeneity.
- Simple to implement and visualize.

**\*\*Disadvantages\*\*:**

- Requires **\*\*pre- and post-treatment\*\*** data for **\*\*both\*\*** groups.
- Relies heavily on the **\*\*parallel trends\*\*** assumption.
- Cannot handle time-varying unobserved confounders easily.

Fill in the following statements about difference in differences so that they are true.

Difference in differences is a causal inference technique that estimates a **treatment effect** by analyzing treatment and control group **trends** over time. We use the control trend to predict a **counterfactual** trend for the treated.

 You got it!

Fill in the R code to create a line plot using ggplot2 that will help assess the parallel trends assumption of difference in differences. The variables in dataset `df` are outcome variable `outcome`, time-point variable `time`, and treatment group status indicator `treat`.

```
import library
library(ggplot2)
line plot of outcome over time

ggplot(data = df, aes(x = time, y = outcome, color = treat, linetype = treat)) +
 geom_line()
```

**treat:time**

**treat\*time**

Click or drag and drop to fill in the blank

Fill in the R code to create a line plot using `ggplot2` that will help assess the parallel trends assumption of difference in differences. The variables in dataset `df` are outcome variable `outcome`, time-point variable `time`, and treatment group status indicator `treat`.

```
import library
library(ggplot2)
line plot of outcome over time

ggplot(data = df, aes(x = time, y = outcome, color = treat, linetype = treat)) +
 geom_line()
```



You got it!

**\*\*Correct Code:\*\***

```
```r
```

```
# import library
```

```
library(ggplot2)
```

```
# line plot of outcome over time
```

```
ggplot(data = df, aes(x = time,
```

```
    y = outcome,
```

```
    color = treat,
```

```
    linetype = treat)) +
```

```
  geom_line()
```

```
```
```

```

```

### Why This Plot?

This visualization is the **\*\*standard way\*\*** to assess the **\*\*parallel trends assumption\*\*** in Difference-in-Differences (DiD).

- **\*\*x-axis (`time`)\*\***: Time periods (pre- and post-treatment).

- **\*\*y-axis (`outcome`)\*\***: The outcome variable.

- **color + linetype (`treat`)**: Differentiates the treatment and control groups.

You want to see two lines that are roughly **parallel before treatment**. Any divergence after treatment suggests a causal effect.

---

### ### Enhanced Version (Recommended)

```
`r
```

```
library(ggplot2)
```

```
ggplot(df, aes(x = time, y = outcome, color = factor(treat), group = treat)) +
```

```
 geom_line(linewidth = 1) +
```

```
 geom_point(size = 3) +
```

```
 labs(title = "Testing Parallel Trends Assumption",
```

```
 x = "Time",
```

```
 y = "Outcome",
```

```
 color = "Group") +
```

```
 theme_minimal()
```

```
...
```

---

### ### Interpretation Guide

- **Parallel pre-treatment lines** → Assumption looks plausible.
- **Diverging lines after treatment** → Evidence of a treatment effect.
- **Non-parallel pre-treatment** → Assumption violated → DiD may be biased.

Which statement below accurately describes the fundamental problem of causal inference?

When randomization is not possible due to ethical or logistical concerns, we cannot estimate causal effects.

Selection bias makes it impossible to estimate causal effects.

The observed outcome is always different from the counterfactual outcome, so we can never estimate the true treatment effect.

We can never observe both potential outcomes, only the one that actually occurs.

This question is about the **fundamental problem of causal inference** in the potential outcomes framework. The correct answer is:

 **"We can never observe both potential outcomes, only the one that actually occurs."**

### Why This Is Correct:

The fundamental problem of causal inference stems from the fact that for any individual or unit, we can only observe **one** of the potential outcomes:

- If someone receives treatment, we observe their outcome under treatment  $Y(1)$ , but we **cannot** observe what would have happened if they hadn't received it  $Y(0)$ .
- If someone doesn't receive treatment, we observe  $Y(0)$ , but we **cannot** observe  $Y(1)$ .

Since causality requires comparing what happened to what *would have* happened (the counterfactual), this missing data problem is fundamentally unresolvable at the individual level.

### Why the Other Options Are Incorrect:

| Option                                                                                                                   | Problem                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "When randomization is not possible... we cannot estimate causal effects"                                                | Too absolute. Observational methods (IV, regress discontinuity, difference-in-differences, propensity estimate causal effects without randomization.              |
| "Selection bias makes it impossible to estimate causal effects"                                                          | Selection bias is a serious challenge, but methods address it. Not "impossible."                                                                                  |
| "The observed outcome is always different from the counterfactual... so we can never estimate the true treatment effect" | Misleading. While we can't observe individual treatment effects, we <i>can</i> estimate <b>average</b> treatment effects across groups using appropriate methods. |

Key Takeaway:

The fundamental problem isn't that causal inference is impossible—it's that **individual-level causal effects are unobservable**. Researchers overcome this by estimating average effects using study designs and statistical methods that make credible assumptions about the missing counterfactuals.

Fill in the R code to create a line plot using ggplot2 that will help assess the parallel trends assumption of difference in differences. The variables in dataset `df` are outcome variable `outcome`, time-point variable `time`, and treatment group status indicator `treat`.

```
import library
library(ggplot2)
line plot of outcome over time

ggplot(data = df, aes(x = time, y = outcome, color = treat, linetype = treat)) +
 geom_line()
```

`treat*time`   `treat:time`

Click or drag and drop to fill in the blank

This question is about the **fundamental problem of causal inference** in the potential outcomes framework. The correct answer is:

✔ **"We can never observe both potential outcomes, only the one that actually occurs."**

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The fundamental problem of causal inference stems from the fact that for any individual or unit, we can only observe **one** of the potential outcomes:

- If someone receives treatment, we observe their outcome under treatment  $Y(1)$ , but we **cannot** observe what would have happened if they hadn't received it  $Y(0)$ .
- If someone doesn't receive treatment, we observe  $Y(0)$ , but we **cannot** observe  $Y(1)$ .

Since causality requires comparing what happened to what *would have* happened (the counterfactual), this missing data problem is fundamentally unresolvable at the individual level.

Why the Other Options Are Incorrect:

| Option                                                                    | Problem                                                                                               |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| "When randomization is not possible... we cannot estimate causal effects" | Too absolute. Observational methods (IV, regress discontinuity, difference-in-differences, propensity |

|                                                                                                                          |                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                          | estimate causal effects without randomization.                                                                                                                    |
| "Selection bias makes it impossible to estimate causal effects"                                                          | Selection bias is a serious challenge, but methods address it. Not "impossible."                                                                                  |
| "The observed outcome is always different from the counterfactual... so we can never estimate the true treatment effect" | Misleading. While we can't observe individual treatment effects, we <i>can</i> estimate <b>average</b> treatment effects across groups using appropriate methods. |

Key Takeaway:

The fundamental problem isn't that causal inference is impossible—it's that **individual-level causal effects are unobservable**. Researchers overcome this by estimating average effects using study designs and statistical methods that make credible assumptions about the missing counterfactuals.

Based on my analysis of the R code shown in the image, the filled-in values appear **correct** for creating a line plot to assess the parallel trends assumption in difference-in-differences (DiD).

Your Code:

```
ggplot(data = df, aes(x = time, y = outcome, color = treat, linetype = treat)) +
 geom_line()
```

| Blank   | Your Answer | Correct? | Reason                                                                 |
|---------|-------------|----------|------------------------------------------------------------------------|
| x =     | time        | ✓        | Time should be on the x-axis to show trends over time                  |
| y =     | outcome     | ✓        | The outcome variable should be on the y-axis                           |
| color = | treat       | ✓        | Different colors for treatment/control groups allows visual comparison |

Why This Works:

To assess **parallel trends**, you need to visually check if the treatment and control groups had similar outcome trajectories **before** the treatment occurred. By plotting:

- Lines separated by `color = treat` (and optionally `linetype = treat` )
- Outcome over time

You can then look at the pre-treatment periods and verify the lines run roughly parallel.

Note on the Extra Options:

The buttons at the bottom ( `treat*time` and `treat:time` ) appear to be unused/distractor options. These would be relevant for specifying an **interaction term** in a regression model (not for a ggplot aesthetic mapping). In model formulas:

- `treat:time` = interaction only
- `treat*time` = main effects + interaction

But for this ggplot visualization, your three filled blanks are correct!

Fill in the correct vocabulary in the following statements:

The first step in causal inference is  , during which we must specify a  (ATE, ATT, ATC, etc.) and determine if we have satisfied the  needed to estimate a treatment effect. The second step in causal inference is  , which involves using a  to estimate the treatment effect.

statistical model

assumptions

causal estimand

estimation

identification

Click or drag and drop to fill in the blank

Based on the standard framework of causal inference, here is the **correct** way to fill in the blanks:

| Blank Position | Correct Word      | Reason                                                                          |
|----------------|-------------------|---------------------------------------------------------------------------------|
| Blank 1        | identification    | First step: defining what ca want to measure and chek identified from the data. |
| Blank 2        | causal estimand   | You specify the target quar ATC, etc.).                                         |
| Blank 3        | assumptions       | You must verify that identif assumptions (unconfounde overlap, etc.) hold.      |
| Blank 4        | estimation        | Second step: actually comp numerical estimate.                                  |
| Blank 5        | statistical model | You use a regression, matc or other model to calculate                          |

Complete Sentence:

"The first step in causal inference is **identification**, during which we must specify a **causal estimand** (ATE, ATT, ATC, etc.) and determine if we have satisfied the **assumptions** needed to



estimate a treatment effect. The second step in causal inference is **estimation**, which involves using a **statistical model** to estimate the treatment effect."

Why This Order Makes Sense:

1. **Identification** → **Estimation** is the canonical two-stage framework in causal inference (Pearl, Imbens, Rubin, etc.)
2. You can't estimate anything until you've **identified** what you're estimating (the estimand) and verified that your data can answer the question (assumptions).
3. Once identified, you move to **estimation** using appropriate **statistical models**.

If your answer matches this sequence, it's **correct!** If you had a different arrangement, feel free to share it and I'll explain what needs adjustment.

True or False: Balance describes how similar the distribution of a variable is between treatment and control groups.

TRUE

FALSE

 **TRUE**

This statement is **correct**.

Why:

**Balance** in causal inference refers to the comparability of treatment and control groups with respect to their covariates (pre-treatment characteristics). Specifically:

- When groups are **balanced**, the distribution of each covariate is similar across treatment and control groups
- When groups are **unbalanced**, there are systematic differences in covariates that could confound the treatment effect estimate

Why Balance Matters:

| Scenario   | Implication                                                                                          |  |
|------------|------------------------------------------------------------------------------------------------------|--|
| Balanced   | Treatment assignment is effectively as-if random; outcome differences can be attributed to treatment |  |
| Unbalanced | Pre-existing differences between groups may confound the estimated treatment effect                  |  |

## Common Contexts:

1. **Randomized Experiments:** Expect balance on average (though random imbalances can occur by chance)
2. **Observational Studies:** Check balance and use techniques to improve it:
  - Propensity score matching
  - Inverse probability weighting
  - Stratification
  - Covariate adjustment in regression

## How to Check Balance:

- Standardized mean differences ( $< 0.1$  is typically considered balanced)
- Visual comparisons (histograms, density plots, love plots)
- Variance ratios

So yes, balance literally describes how similar the distribution of variables (covariates) is between treatment and control groups.

Take a look at the output below generated by the `bal.tab()` function from the `cobalt` package. Does this output suggest good balance or poor balance? Why?

```
Balance Measures
 Type Diff.Un V.Ratio.Un
variable_1 Contin. 0.2141 1.5202
variable_2 Contin. -0.4405 0.6732

Sample sizes
 Control Treated
All 120 46
```

Good balance because one standardized mean difference is positive while the other is negative.

Good balance because both of the variance ratios are positive.

Poor balance because the control group has a sample size of 120 while the treatment group only has 46.

Poor balance because the standardized mean differences are not close to 0 and the variance ratios are different from 1.

We make four main assumptions about the sample in instrumental variable estimation. Match the definitions below to the correct assumption.

1. Relevance:

2. Exchangeability:

3. Exclusion:

4. Monotonicity:

There are no confounding variables which influence both the instrument AND the outcome.

The instrument has a causal effect on the treatment received.

The instrument affects the outcome ONLY indirectly through the treatment received.

There are no "defiers" in the sample.

Click or drag and drop to fill in the blank

We make four main assumptions about the sample in instrumental variable estimation. Match the definitions below to the correct assumption.

1. Relevance:

2. Exchangeability:

3. Exclusion:

4. Monotonicity:

Click or drag and drop to fill in the blank

Here's the **correct way** to match these instrumental variable (IV) estimation assumptions with their definitions:

| Assumption         | Correct Definition                                                                      |  |
|--------------------|-----------------------------------------------------------------------------------------|--|
| 1. Relevance       | The instrument has a causal effect on the treatment received.                           |  |
| 2. Exchangeability | There are no confounding variables which influence both the instrument AND the outcome. |  |
| 3. Exclusion       | The instrument affects the outcome ONLY indirectly through the treatment received.      |  |

**4. Monotonicity**

There are no "defiers" in the sample.

## Quick Summary:

| # | Assumption      | Match To                                                                                  |  |
|---|-----------------|-------------------------------------------------------------------------------------------|--|
| 1 | Relevance       | ✓ The instrument has a causal effect on the treatment received.                           |  |
| 2 | Exchangeability | ✓ There are no confounding variables which influence both the instrument AND the outcome. |  |
| 3 | Exclusion       | ✓ The instrument affects the outcome ONLY indirectly through the treatment received.      |  |
| 4 | Monotonicity    | ✓ There are no "defiers" in the sample.                                                   |  |

## Mnemonic for Remembering:

- **Relevance** = Instrument must be *relevant* to treatment (correlated/causal)
- **Exchangeability** = Instrument is *exchangeable* across subgroups (no confounders)
- **Exclusion** = Instrument is *excluded* from affecting outcome directly (only through treatment)
- **Monotonicity** = Everyone moves in one direction (no *defiers*)

This is a cleaner version of the earlier question where the definitions were swapped! Make sure you drag each definition to its matching box according to the table above.

Regression discontinuity design uses only a subset of the full data. Which of the following statements about the causal treatment effect we can estimate using RDD is true?

Results of a regression discontinuity design (RDD) analysis are easily generalizable because RDD estimates the local average treatment effect (LATE).

We can estimate the average treatment effect (ATE) for the full data.

We cannot estimate any causal treatment effect.

We can only estimate the treatment effect for the subset of the full data, known as the local average treatment effect (LATE).

✓

Correct Answer:

"We can only estimate the treatment effect for the subset of the full data, known as the local average treatment effect (LATE)."

Why This Is Correct:

Regression Discontinuity Design (RDD) exploits a **cutoff/threshold** on a running variable to assign treatment. For example:

- Students scoring  $\geq 60$  on a test receive tutoring (treatment), those below do not
- Voters in districts with  $>50\%$  minority population get special provisions

The key insight: RDD compares units just above and just below the cutoff. Because units very close to the threshold are essentially randomly assigned (small differences in the running variable determine treatment status), we can identify a **causal effect**.

However, this effect is only identifiable for individuals **near the cutoff** — hence "**local**" average treatment effect.

Why the Other Options Are Wrong:

| Option                                                           | Problem                                                                                                         |  |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--|
| "Results... are easily generalizable because RDD estimates LATE" | Contradictory! LATE is <i>local</i> , meaning it has <b>limited generalizability</b> to the broader population. |  |
| "We can estimate ATE for the full data"                          | False. ATE requires knowing effects for everyone. RDD only gives us info near the cutoff.                       |  |
| "We cannot estimate any causal treatment effect"                 | False. RDD is specifically designed to identify <b>causal</b> effects — that's its strength!                    |  |

Key Properties of RDD:

| Property   | Description                                                                |  |
|------------|----------------------------------------------------------------------------|--|
| Identifies | Local Average Treatment Effect (LATE) near the cutoff                      |  |
| Strength   | Internal validity is high (quasi-experimental)                             |  |
| Limitation | External validity is limited (hard to generalize beyond the cutoff region) |  |

|            |                                                                          |  |
|------------|--------------------------------------------------------------------------|--|
| Assumption | Units cannot perfectly manipulate the running variable around the cutoff |  |
|------------|--------------------------------------------------------------------------|--|

Bottom Line:

RDD gives us **credible causal estimates**, but only for the "marginally treated" — those right at the decision boundary. Generalizing to the full population requires additional assumptions or data.

Which of the following coefficients represents the ATT in the Difference in Difference regression output?

|             |           |          |            |
|-------------|-----------|----------|------------|
| (Intercept) | treat     | time     | treat:time |
| 672.81456   | 35.831336 | 27.18544 | 3.4168664  |

treat

treat:time

(Intercept)

time

 **Correct Answer:**

"treat:time"

Why This Is Correct:

In a standard Difference-in-Differences (DiD) regression, the model looks like this:

$Y = \beta_0 + \beta_1(\text{treat}) + \beta_2(\text{time}) + \beta_3(\text{treat} \times \text{time}) + \epsilon$

| Coefficient           | Represents                                 | Value from Output                                                                               |  |
|-----------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------|--|
| (Intercept) $\beta_0$ | Control group baseline (pre-period)        | 672.81456                                                                                       |  |
| treat $\beta_1$       | Treatment-control difference in pre-period | 35.831336                                                                                       |  |
| time $\beta_2$        | Common time trend (both groups)            | 27.18544                                                                                        |  |
| treat:time $\beta_3$  | ATT (DiD estimator)                        | 3.4168664  |  |

The **interaction term (treat:time)** captures the *additional* change in outcomes for the treatment group relative to the control group after treatment — this is exactly the **Average Treatment Effect on the Treated (ATT)**.

## Visual Breakdown of the DiD Logic:

|                 | Pre-Treatment       | Post-Treatment                             | Difference        |  |
|-----------------|---------------------|--------------------------------------------|-------------------|--|
| Control Group   | (Intercept)         | (Intercept) + time                         | time              |  |
| Treatment Group | (Intercept) + treat | (Intercept) + treat + time<br>+ treat:time | time + treat:time |  |
| Difference      | treat               | treat + treat:time                         | treat:time ← ATT  |  |

The DiD estimator subtracts the control group's change from the treatment group's change: (treatment post–treatment pre)–(control post–control pre)=treat:time

## Bottom Line:

- **treat:time** = The causal effect of the treatment on those who received it (ATT) = **3.4168664**
- All other coefficients are nuisance parameters (baseline, group differences, time trends)

| Z       | $Y^0$ | $Y^1$ | Y   | $Y^1 - Y^0$ |
|---------|-------|-------|-----|-------------|
| 1       | 5     | 7     | 7   | 2           |
| 1       | 4     | 5     | 5   | 1           |
| 0       | 6     | 10    | 6   | 4           |
| 1       | 2     | 5     | 5   | 3           |
| 0       | 8     | 8     | 8   | 0           |
| AVERAGE | 5     | 7     | 6.2 | 2           |

The table above shows treatment and outcome data from 5 individuals using the standard notation for treatment assignment (Z), potential outcomes ( $Y^0$ ,  $Y^1$ ), and observed outcome (Y). Which column contains the individual treatment effect (ITE) and what is the true average treatment effect (ATE)?

$$ITE = Y^1, ATE = 7$$

$$ITE = Y, ATE = 6.2$$

$$ITE = Y^1 - Y^0, ATE = 2$$

$$ITE = Y^0, ATE = 6$$

**\*\*Correct Answer:\*\***

$$**ITE = Y^1 - Y^0, ATE = 2**$$

---

### ### Explanation

In the **potential outcomes framework**:

- **$Y^0$** : Potential outcome if the individual had **not** received treatment.
- **$Y^1$** : Potential outcome if the individual **had** received treatment.
- **$Y$** : The **observed** outcome (only one of  $Y^0$  or  $Y^1$  is observed for each person, depending on  $Z$ ).

The **Individual Treatment Effect (ITE)** for each person is the difference in their potential outcomes:

\$\$

$$\text{ITE}_i = Y^1_i - Y^0_i$$

\$\$

This is exactly the column labeled  **$Y^1 - Y^0$** .

---

### ### Average Treatment Effect (ATE)

The **true Average Treatment Effect (ATE)** is the average of all individual treatment effects across the population:

\$\$

$$\text{ATE} = \frac{1}{N} \sum_{i=1}^N (Y^1_i - Y^0_i)$$

\$\$

You can see this directly from the bottom row of the table (average of the last column = 2).



---

### ### Why the Other Options Are Incorrect

- **$Y^1$**  alone is just the potential outcome under treatment, not the effect.
- **$Y$**  (observed outcome) mixes treated and untreated outcomes — it does **not** give the causal effect.
- **$Y^0$**  is just the control potential outcome.

---

### ### Key Takeaway

The fundamental challenge in causal inference is that we can **never** observe both  $Y^0$  and  $Y^1$  for the same individual at the same time. This is known as the **Fundamental Problem of Causal Inference**. We must use statistical methods (RCTs, IV, RDD, DiD, Matching, etc.) to estimate the ATE.

Would you like a simulated example in R/Python showing how we estimate the ATE when we only observe  $Y$ ?

Fill in the following statements about difference in differences so that they are true.

Difference in differences is a causal inference technique that estimates a **treatment effect** by analyzing treatment and control group **trends** over time. We use the control trend to predict a **counterfactual** trend for the treated.

**observed** **linear effect**

Click or drag and drop to fill in the blank

Which of the following is NOT an assumption for DID?

Data must be normally distributed.

The groups within the data need to have parallel trends over time.

Conditional exchangeability

Treatment for the treated was not determined by the value of the outcome variable before treatment assignment.

 **Correct Answer:**

"Data must be normally distributed."

**Why This Is NOT a Core DiD Assumption:**

Normality of the data distribution is **not required** for the identification of the causal effect in Difference-in-Differences. While normality may be assumed for classical hypothesis testing (t-tests, confidence intervals with small samples), modern DiD applications typically use:

- Robust standard errors
- Clustered standard errors
- Bootstrap methods
- Large-sample asymptotic theory

These approaches do not require normally distributed data.

**Why the Other Options ARE DiD Assumptions:**

| Option                                              | Why It's an Assumption                                                                                                                                                                                                                                     |  |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| "Parallel trends over time"                         |  <b>Core assumption:</b> Without treatment, treatment and control groups would have followed the same outcome trajectory. This is the foundation of DiD identification. |  |
| "Conditional exchangeability"                       |  <b>Important assumption:</b> Conditional on covariates, there are no unobserved confounders that vary by group and time. Related to parallel trends.                   |  |
| "Treatment not determined by pre-treatment outcome" |  <b>Valid assumption:</b> Prevents anticipation effects or selection based on prior outcomes (no "good news/bad news" treatment assignment).                            |  |

Summary Table:

| Statement                               | Is It a DiD Assumption? |  |
|-----------------------------------------|-------------------------|--|
| Data must be normally distributed       | ✗ NOT                   |  |
| Parallel trends over time               | ✓ Yes                   |  |
| Conditional exchangeability             | ✓ Yes                   |  |
| Treatment not determined by pre-outcome | ✓ Yes                   |  |

Bottom Line:

The correct answer is **"Data must be normally distributed"** because DiD identifies causal effects through the structure of the design (two groups × two time periods), not through distributional assumptions about the error terms or outcomes.

Which of the following accurately describes the purpose and stages of two-stage least squares (2SLS) regression for instrumental variable (IV) estimation?

2SLS is used to estimate the instrument effect in IV estimation. The first stage predicts the outcome from the treatment received. The second state uses the predicted outcome to estimate the instrument.

2SLS is used to estimate the causal treatment effect in IV estimation. The first stage predicts the treatment received from the instrument. The second stage uses the predicted treatment received in the first stage to predict the outcome.

2SLS is used to estimate the causal treatment effect in IV estimation. The first stage predicts the instrument from the treatment received. The second stage uses the instrument predicted in the first stage to predict the outcome.

✓ Correct Answer: Option 2

"2SLS is used to estimate the causal treatment effect in IV estimation. The first stage predicts the treatment received from the instrument. The second stage uses the predicted treatment received in the first stage to predict the outcome."

Why Option 2 Is Correct:

Two-Stage Least Squares (2SLS) works as follows:

| Stage       | What Happens                                                                        | Regression                |  |
|-------------|-------------------------------------------------------------------------------------|---------------------------|--|
| First Stage | Regress <b>treatment</b> on the <b>instrument</b> to get predicted treatment values | $T^{\wedge}=\pi_0+\pi_1Z$ |  |

|                     |                                                                             |                                  |  |
|---------------------|-----------------------------------------------------------------------------|----------------------------------|--|
| <b>Second Stage</b> | Regress <b>outcome</b> on <b>predicted treatment</b> (not actual treatment) | $Y = \beta_0 + \beta_1 T^\wedge$ |  |
|---------------------|-----------------------------------------------------------------------------|----------------------------------|--|

The key idea: by replacing actual treatment with **predicted treatment** (which is driven only by the instrument), we isolate the variation in treatment that is **uncorrelated with confounders**, giving us a causal estimate.

## Why the Other Options Are Wrong:

| Option          | Problem                                                                                                                                                                                  |  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Option 1</b> | Says the first stage predicts the <i>outcome</i> (wrong — it predicts <i>treatment</i> ) and the goal is to estimate the "instrument effect" (wrong — it's the <i>treatment effect</i> ) |  |
| <b>Option 3</b> | Reverses the first stage: predicts the <i>instrument</i> from <i>treatment</i> (backwards — should be treatment from instrument)                                                         |  |

## Simple Memory Aid:

$$Z \rightarrow \hat{T} \rightarrow Y$$

Instrument predicts treatment → predicted treatment predicts outcome → causal effect!

The coefficient  $\beta_1$  on  $T^\wedge$  in the second stage is your **causal treatment effect** (specifically the LATE under the IV assumptions).

True or False: the primary disadvantages of instrumental variable (IV) estimation are that the causal estimand is not generalizable and that finding a good instrument is difficult.

False

True

## ✓ Correct Answer: True

This statement is **accurate**. Both points mentioned are widely recognized as the primary disadvantages of instrumental variable (IV) estimation.

## 1. Causal Estimand Is Not Generalizable

| Issue             | Explanation                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| What IV Estimates | Local Average Treatment Effect (LATE)                                                                                           |
| Who It Applies To | Only "compliers" — individuals whose treatment status actually changed by the instrument                                        |
| Who Is Excluded   | Never-takers (never take treatment regardless of instrument) and Always-takers (always take treatment regardless of instrument) |
| Implication       | Results cannot be easily generalized to the full population                                                                     |

Unlike methods that can estimate ATE (Average Treatment Effect) for everyone, IV only tells us about the marginal subgroup induced to change behavior by the instrument.

## 2. Finding a Good Instrument Is Difficult

A valid instrument must satisfy **all three** conditions:

| Condition             | Challenge                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------|
| Relevance             | Must strongly predict the treatment (weak instrument leads to biased estimates)             |
| Exclusion Restriction | Cannot directly affect the outcome (untestable assumption; must be justified theoretically) |
| Exchangeability       | No confounders between instrument and outcome (untestable)                                  |

**Reality:** Very few real-world variables convincingly satisfy all three. Common examples (weather, distance, policy shocks, genetic variants) often face skepticism about their validity.

### Additional Disadvantages (Bonus):

- **Lower statistical power** compared to OLS (due to two-stage estimation)
- **Sensitive to weak instruments** (can produce biased/inconsistent estimates)
- **Hard to defend exclusion restriction** (often relies on theoretical arguments rather than empirical proof)

### Bottom Line:

The statement correctly identifies the two biggest practical limitations of IV: **limited generalizability** (LATE only) and **difficulty finding valid instruments**. These are standard criticisms taught in causal inference courses.

If filled in correctly, the code below performs inverse probability of treatment weighting (IPTW). Fill in the blanks in the code below with the correct arguments.

```
import library
library(WeightIt)
perform IPTW in weightit object
weightit(
 formula = ,
 data = test_data,
 estimand = ,
 method =
)
```

Click or drag and drop to fill in the blank

If filled in correctly, the code below performs inverse probability of treatment weighting (IPTW). Fill in the blanks in the code below with the correct arguments.

```
import library
library(WeightIt)
perform IPTW in weightit object
weightit(
 formula = ,
 data = test_data,
 estimand = ,
 method =
)
```

Click or drag and drop to fill in the blank

Based on the structure of the `WeightIt` package in R and the principles of Inverse Probability of Treatment Weighting (IPTW), here is the correct way to fill in the blanks:

 **Correct Answers:**

| Parameter | Correct Argument                     |  |
|-----------|--------------------------------------|--|
| formula   | <code>treatment ~ var1 + var2</code> |  |

|          |       |  |
|----------|-------|--|
| estimand | "ATE" |  |
| method   | "ps"  |  |

Why These Are Correct:

| Blank                                          | Explanation                                                                                                                                                                     |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>formula = treatment ~ var1 + var2</code> | In IPTW, we model <b>treatment assignment</b> as a function of covariates to estimate the propensity score. We do not model the outcome in this formula (that would be wrong!). |
| <code>estimand = "ATE"</code>                  | Specifies we want to estimate the Average Treatment Effect on the full population (alternatives include <code>"ATT"</code> for Treatment Effect on the Treated).                |
| <code>method = "ps"</code>                     | Stands for <b>propensity score</b> weighting, which is the standard method for IPTW.                                                                                            |

Complete Code:

```
library(WeightIt)

weightit(
 formula = treatment ~ var1 + var2,
 data = test_data,
 estimand = "ATE",
 method = "ps"
)
```

Why the Other Options Are Wrong:

| Option                                         | Why Incorrect                                                                                                                                       |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>outcome ~ treatment + var1 + var2</code> | This is an outcome regression formula, not a propensity score formula. For IPTW, we need to model treatment assignment as a function of covariates. |
| <code>"SMD"</code>                             | Standardized Mean Difference is a diagnostic measure of balance, not an estimation method.                                                          |
| <code>"optimal"</code>                         | This is a valid method in WeightIt, but <code>"ps"</code> is the basic IPTW with propensity scores.                                                 |

Quick Reminder:

For IPTW/propensity score weighting:

- 1. **Model treatment** → **covariates** (not outcome!)
- 2. **Calculate weights** from predicted probabilities
- 3. **Estimate effect** using weighted regression

The correct answer is:

- **formula:** `treatment ~ var1 + var2`
- **estimand:** `"ATE"`
- **method:** `"ps"`

There are many different ways to define relationships between variables in causal inference. Which of the following is true about the difference between association and causation?

Association is a relationship without a specified direction, while a causal relationship does have a specified direction.

Association is a linear relationship while causation implies a non-linear relationship.

A causal relationship does not specify the strength or pattern of the relationship, while an association does specify the strength or pattern of the relationship.

## Correct Answer:

"Association is a relationship without a specified direction, while a causal relationship does have a specified direction."

### Why This Is Correct:

| Concept     | Directionality         | Example                                                                |
|-------------|------------------------|------------------------------------------------------------------------|
| Association | No specified direction | Ice cream sales ↔ Drownin<br>correlate, but neither cause<br>directly) |
| Causation   | Clear direction        | Smoking → Lung cancer (s<br>causes cancer, not the rev                 |

**Association** tells us variables move together but doesn't tell us:

- Which one influences the other
- Whether there's a third common cause (confounding)

**Causation** explicitly states:

- X affects Y (direction matters!)
- Changing X will change Y (intervention perspective)



# Why the Other Options Are Wrong:

| Option                                                                | Problem                                                                                                                                          |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| "Association is linear while causation is non-linear"                 | False! Both association and causation can be linear. Linearity is unrelated to the distinction.                                                  |
| "Causal relationship doesn't specify strength while association does" | False! It's actually reversed—causal estimates (like RCTs) explicitly quantify strength. Correlations also quantify strength but both can do so. |

## Key Takeaway:

The fundamental difference is about **direction and intervention**:

| Feature      | Association                           | Causation                         |  |
|--------------|---------------------------------------|-----------------------------------|--|
| Direction    | Bidirectional/undefined               | Unidirectional (X→Y)              |  |
| Intervention | Observation only                      | Predicts effect of manipulating X |  |
| Confounding  | Susceptible to spurious relationships | Must account for confounders      |  |

**Correlation does not imply causation** because association alone doesn't tell you which variable affects the other—or if a hidden third factor drives both!

The code block below shows IPTW procedures performed using a mock dataset. Select the line of code with a regression model that could be used in the final step of a propensity score analysis to get an estimate of the causal treatment effect.

```
import library
library(WeightIt)
perform IPTW in weightit object
wt_iptw <- weightit(
 formula = treat ~ var1 + var2,
 data = test_data,
 estimand = "ATT",
 method = "ps"
)
```

```
glm(outcome ~ treat + var1 + var2, data = test_data, weights = wt_iptw$weights)
```

```
glm(outcome ~ treat + var1 + var2, data = test_data)
```

```
glm(treat ~ var1 + var2, data = test_data, weights = wt_iptw$weights)
```

```
glm(outcome ~ treat + var1 + var2, data = test_data, weights = wt_iprw$weights)
```

## Why This Line?

This is the **final step** in Inverse Probability of Treatment Weighting (IPTW) using the `WeightIt` package.

## IPTW Workflow Summary:

1. **Estimate propensity scores** and compute weights (`weightit()`).
2. **Fit a weighted regression** on the **outcome**, using the treatment indicator as the main predictor and the IPTW weights.

The weighted regression adjusts for confounding by giving more weight to individuals who were underrepresented based on their covariates.

---

## Interpretation

- The coefficient on **treat** in this weighted model gives the estimated **causal treatment effect** (usually the ATE).
- Including `var1 + var2` helps further reduce bias (doubly robust approach).

---

## Complete Example

R

```
library(WeightIt)

Step 1: Estimate weights
wt_iprw <- weightit(treat ~ var1 + var2,
 data = test_data,
 estimand = "ATT",
 method = "ps")

Step 2: Weighted outcome model (FINAL STEP)
model <- glm(outcome ~ treat + var1 + var2,
 data = test_data,
 weights = wt_iprw$weights)

summary(model)
```

---

## Why the Other Options Are Wrong

- `glm(outcome ~ treat + var1 + var2)` → Unweighted model (ignores propensity score adjustment).
- `glm(treat ~ var1 + var2, weights = ...)` → This is modeling the **propensity score** itself (already done in the first step).

Pretend that you have access to a dataset called `student_data` about elementary student performance containing the following variables:

- `t-scores` — standardized test score from 1 to 100.
- `private_school` — indicates what type of school the student attended (1 = private school; 0 = public school).
- `n_siblings` — number of siblings living in the household.
- `hh_income` — household income in thousands.

Suppose you want to determine the effect of attending private school on test scores by conducting a propensity score analysis. Complete the code below to fit a propensity score model that estimates the probability of being in the treatment group (attending private school) given household income.

```
glm(
 formula = ~ ,
 data = student_data,
 family =)
```

private\_school

"logistic"

n\_siblings

hh\_income

"binomial"

✔ **Correct Answers:**

| Blank                 | Correct Argument |
|-----------------------|------------------|
| Left side of formula  | private_school   |
| Right side of formula | hh_income        |
| family                | "binomial"       |

**Complete Code:**

```
glm(
 formula = private_school ~ hh_income,
 data = student_data,
 family = "binomial")
```

**Why These Are Correct:**

| Blank                 | Explanation                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| private_school (left) | This is the <b>dependent variable</b> (treatment assignment modeling the probability of being in the treatment school = 1 vs. public = 0). |

|                                |                                                                                                                                                       |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hh_income</code> (right) | This is the <b>predictor/covariate</b> . The question specifies "household income"—so only this variable goes into the formula.                       |
| <code>"binomial"</code>        | This specifies the distribution family. For binary outcomes we use the binomial family, which performs logistic regression using <code>glm()</code> . |

## Why NOT the Other Options?

| Option                  | Why Incorrect                                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>"logistic"</code> | While this describes the <i>type</i> of regression, R's <code>glm()</code> requires <code>"binomial"</code> as the actual parameter for the distribution family.    |
| <code>n_siblings</code> | The question specifically says "given household income, not siblings." Including it would violate the problem constraints.                                          |
| <code>t_scores</code>   | This is the <b>outcome variable</b> we care about later. It is a covariate for the propensity model. The propensity model predicts <b>treatment</b> , not outcomes! |

## Quick Tip for Propensity Scores:

The formula always follows:

```
treatment ~ covariates
```

And for binary treatment:

```
family = "binomial"
```

This creates the propensity scores needed for matching, weighting, or stratification in your subsequent causal analysis!

In regression discontinuity design (RDD), the causal treatment effect is estimated in a subset of the full sample. Which term is used to describe the interval of forcing variable values, determined either empirically or algorithmically, that is used to subset the data?

confidence interval

cutpoint

local average treatment effect (LATE)

bandwidth

 **Correct Answer: bandwidth**

**Why "Bandwidth" Is Correct:**

In Regression Discontinuity Design (RDD), the **bandwidth** refers to the interval or window of values around the **cutpoint** (threshold) that is used to subset the data for estimation.

| Term      | What It Means                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------|
| Bandwidth | The range of forcing variable values included in the analysis (e.g., ±10 points around the cutpoint) |
| Purpose   | To focus on observations close to the threshold where treatment assignment is quasi-random           |
| Selection | Can be determined empirically (cross-validation, placebo tests) or algorithmically                   |

**Example:** If the cutpoint is a test score of 60 and you use a bandwidth of 10, you'd include students with scores from 50 to 70 in your RDD analysis.

**Why the Other Options Are Wrong:**

| Option              | Problem                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------|
| Confidence interval | Used to express uncertainty around an estimate (e.g., 95% CI), not for selecting which data to include          |
| Cutpoint            | This is the <b>single threshold value</b> itself (e.g., 60), not the interval around it                         |
| LATE                | Refers to the type of causal effect estimated (Local Average Treatment Effect), not a data-subsetting parameter |

**Key RDD Concepts:**

| Concept            | Description                                                                     |
|--------------------|---------------------------------------------------------------------------------|
| Forcing variable   | The continuous variable that determines treatment assignment (e.g., test score) |
| Cutpoint/Threshold | The value where treatment switches (e.g., score ≥ 60)                           |
| Bandwidth          | The window around the cutpoint used for estimation                              |
| Local effect       | RDD estimates effects only for units near the threshold (within the bandwidth)  |

## Bottom Line:

The **bandwidth** determines how much data you include in your RDD analysis—narrower bandwidths give more precise causal identification but less statistical power, while wider bandwidths increase power but introduce more bias. Choosing the right bandwidth is a critical step in RDD!

Suppose that we have access to a dataframe named `df` that contains several variables:

- `forcing_var`: a forcing variable;
- `outcome_var`: an outcome variable; and
- `trt_group`: a treatment group variable.

Complete the code below to produce a scatter plot of the data in `df` that could be used to check the assumptions for a regression discontinuity design analysis.

```
import library
library(ggplot2)
RDD plot

ggplot(data = df, aes(x = forcing_var, y = outcome_var, color = trt_group, shape
= trt_group)) +
 geom_point()
```

`geom_line()`

Click or drag and drop to fill in the blank

### Completed Code:

R

```
import library
library(ggplot2)

RDD plot
ggplot(data = df, aes(x = forcing_var,
 y = outcome_var,
 color = trt_group,
 shape = trt_group)) +
 geom_point()
```

### Correct matches for the blanks:

- `x = forcing_var`
- `y = outcome_var`
- `color = trt_group`
- Layer: `geom_point()`

## Why This Plot? (RDD Diagnostic Visualization)

In **Regression Discontinuity Design (RDD)**, the core idea is that treatment assignment is determined by a cutoff in the **forcing/running variable** (e.g., test score, age, income). We assume that individuals just above and below the cutoff are otherwise similar (continuity of potential outcomes).

This scatter plot is the **standard visual diagnostic** to:

1. Check for a **discontinuity** (jump) in the outcome at the cutoff → evidence of treatment effect.
2. Assess the functional form (linear? nonlinear?) on both sides.
3. Visualize how the treatment group (trt\_group) behaves around the threshold.

**Key assumptions this helps check:**

- **Continuity of the forcing variable density** (no manipulation around cutoff).
- **Continuity of covariates** (no jumps in other variables).
- Functional form / bandwidth choice.

## Enhanced Version (Recommended for Better Insight)

While the basic `geom_point()` completes the question, here's a more informative version used in practice:

R

```
library(ggplot2)

ggplot(df, aes(x = forcing_var, y = outcome_var, color = trt_group)) +
 geom_point(alpha = 0.6, size = 2) + # Scatter
 geom_smooth(method = "lm", se = FALSE, linewidth = 1) + # Linear fit on
each side
 geom_vline(xintercept = cutoff_value, linetype = "dashed", color = "red")
+ # Cutoff line
 labs(title = "Regression Discontinuity Design - Visual Check",
 x = "Forcing Variable (e.g., Score)",
 y = "Outcome Variable",
 color = "Treatment Group") +
 theme_minimal()
```

**Python equivalent** (using seaborn + matplotlib, very common in Data Science):

Python

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.scatterplot(data=df, x='forcing_var', y='outcome_var',
 hue='trt_group', style='trt_group', alpha=0.7)
sns.lmplot(data=df, x='forcing_var', y='outcome_var',
 hue='trt_group', ci=None, scatter=False)
plt.axvline(x=cutoff_value, color='red', linestyle='--')
```

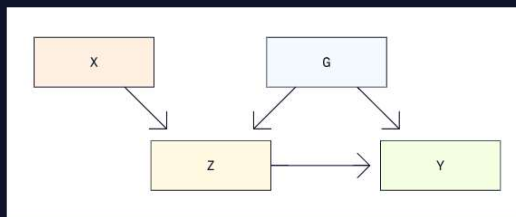
```
plt.title('RDD Scatter Plot for Assumption Checking')
plt.show()
```

## Practical Guidance

- **When to use RDD:** Sharp cutoff rules (e.g., scholarship if score > 80, policy implemented at a date, etc.).
- **Next steps after this plot:**
  - A. McCrary density test (check manipulation).
  - B. Formal RDD estimation (rdrobust package in R, or rdd / rd in Python).
  - C. Robustness checks with different bandwidths and polynomial orders.

The diagram below shows the relationship between a hypothetical treatment Z and an outcome variable Y.

True or False: The variable X is a confounding variable.

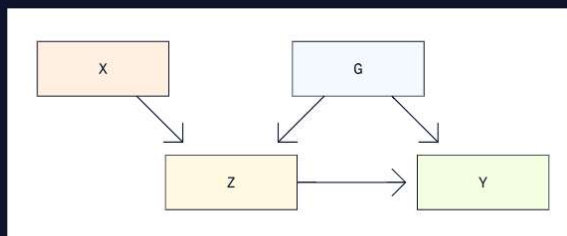


False

True

The diagram below shows the relationship between a hypothetical treatment Z and an outcome variable Y.

True or False: The variable X is a confounding variable.



False

True



Which of the following is NOT true about regression discontinuity design (RDD)?

When a forcing variable cutpoint is exact, the RDD is known as a sharp RDD. When the cutpoint is not exact, the RDD is known as a fuzzy RDD.

In RDD, the forcing variable is NOT related to the outcome variable of interest.

RDD may be used for causal inference when the treatment is assigned to observations that fall above or below a specific cutpoint.

In RDD, a forcing variable is a continuous variable with a cutpoint that determines the treatment group assignment.

Which of the following is NOT true about regression discontinuity design (RDD)?

When a forcing variable cutpoint is exact, the RDD is known as a sharp RDD. When the cutpoint is not exact, the RDD is known as a fuzzy RDD.

In RDD, the forcing variable is NOT related to the outcome variable of interest.

RDD may be used for causal inference when the treatment is assigned to observations that fall above or below a specific cutpoint.

In RDD, a forcing variable is a continuous variable with a cutpoint that determines the treatment group assignment.

✓ **Correct Answer: "In RDD, the forcing variable is NOT related to the outcome variable of interest."**

This statement is **FALSE**.

### Why This Is Not True:

In RDD, the forcing variable **IS related** to the outcome variable. In fact, the entire premise of RDD relies on this relationship:

| Reality                    | Explanation                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forcing variable → Outcome | The forcing variable typically has a continuous relationship with the outcome (e.g., higher test scores generally lead to higher future performance) |

|                     |                                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| What RDD Identifies | The <b>jump/discontinuity</b> in the outcome at the cutoff <i>beyond</i> the smooth relationship with the forcing variable             |
| Key Insight         | Units just above and below the cutoff are similar in all respects except treatment assignment, allowing us to isolate treatment effect |

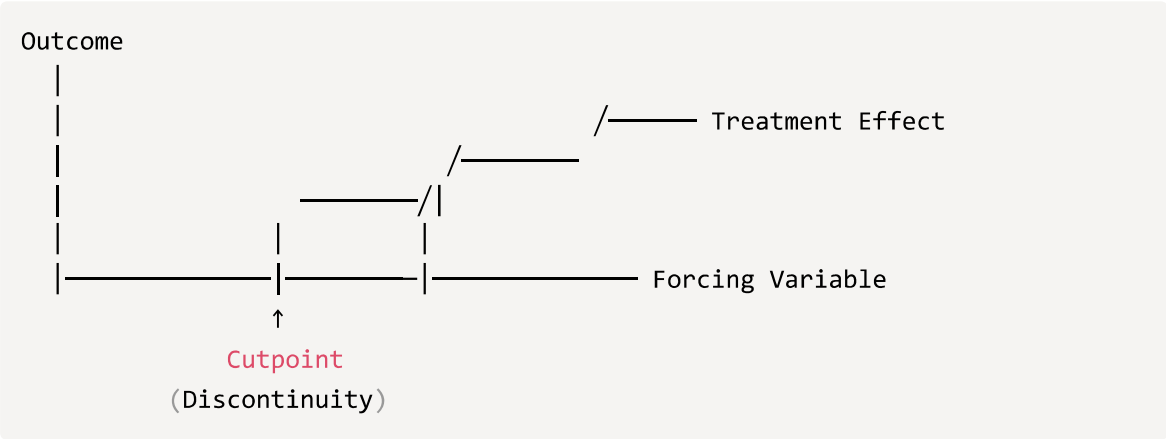
Why the Other Statements Are True:

|                                                |                                                                                                                                                                                                                |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option                                         | Why True                                                                                                                                                                                                       |
| "Sharp vs. Fuzzy RDD"                          | <input checked="" type="checkbox"/> <b>Sharp:</b> Treatment deterministically assigned at cutoff (e.g., age ≥ 18). <b>Fuzzy:</b> Treatment probability changes at cutoff (e.g., age ≥ 18 with non-compliance). |
| "Treatment assigned above/below cutpoint"      | <input checked="" type="checkbox"/> This is the basic mechanism of RDD — eligibility determined by crossing a threshold.                                                                                       |
| "Forcing variable is continuous with cutpoint" | <input checked="" type="checkbox"/> The forcing (or running) variable must be continuous at the cutoff to determine treatment assignment at a specific point.                                                  |

Key RDD Concepts:

|                           |                                                                            |
|---------------------------|----------------------------------------------------------------------------|
| Concept                   | Description                                                                |
| Forcing Variable          | Continuous variable that determines treatment (e.g., age, income)          |
| Cutpoint                  | Threshold where treatment switches (e.g., score ≥ 70)                      |
| Relationship with Outcome | Expected to be smooth, except for the treatment-induced jump at the cutoff |
| Identification Strategy   | Compare outcomes just above vs. just below the cutoff                      |

Visual Understanding:



The forcing variable clearly relates to the outcome (smooth trend), but there's a **discontinuity** at the cutoff that represents the treatment effect!

Bottom Line:

The false statement is: "In RDD, the forcing variable is NOT related to the outcome variable of interest."

The forcing variable *must* relate to the outcome—RDD isolates the treatment effect by comparing units near the cutoff while controlling for this continuous relationship.

Imagine that we have access to a dataset named `df` which contains several variables:

- `outcome`: the outcome of interest;
- `treat`: the treatment variable;
- `instrument`: the instrument variable;

Fill in the code below to fit an appropriate two-stage least squares (2SLS) regression model for instrumental variable estimation.

```
import library
library(AER)
2SLS model
ivreg(
 formula = ~ | ,
 data =
)
```

outcome

df

treat

instrument

Click or drag and drop to fill in the blank

✔ Correct Answers:

| Blank Position | Correct Argument |
|----------------|------------------|
| Left of ~      | outcome          |
| Right of ~     | treat            |
| **After `      | `**              |
| data           | df               |

## Complete Code:

```
import library
library(AER)
2SLS model
ivreg(
 formula = outcome ~ treat | instrument,
 data = df
)
```

## Why This Syntax Is Correct:

The `ivreg()` function from the **AER** package uses a specific formula notation for instrumental variables:

| Formula Part                 | Meaning                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>outcome ~ treat</code> | Left side of <code>~</code> : dependent variable<br>Right side of <code>~</code> : endogenous treatment variable |
| <code>**`</code>             | instrument`**                                                                                                    |
| <code>data = df</code>       | The dataset containing all variables                                                                             |

## How 2SLS Works Here:

| Stage        | Equation                                                                            | What Happens               |
|--------------|-------------------------------------------------------------------------------------|----------------------------|
| First Stage  | <code>treat = <math>\pi_0</math> + <math>\pi_1</math>(instrument)</code>            | Predict treatment from the |
| Second Stage | <code>outcome = <math>\beta_0</math> + <math>\beta_1</math>(treat_predicted)</code> | Regress outcome on predi   |

The coefficient  $\beta_1$  on `treat` gives you the **causal treatment effect** (LATE) identified by the instrument.

## Quick Memory Aid:

```
outcome ~ treatment || instrument
```

Read it as: "outcome depends on treatment, where treatment is instrumented by..."

This is the standard notation for 2SLS in R's `ivreg()` function!

Imagine that we have access to a dataset named `df` which contains several variables:

- `outcome`: the outcome of interest;
- `treat`: the treatment variable;
- `instrument`: the instrument variable;

Fill in the code below to fit an appropriate two-stage least squares (2SLS) regression model for instrumental variable estimation.

```
import library
library(AER)
2SLS model
ivreg(
 formula = outcome ~ treat | instrument ,
 data = df
)
```

Click or drag and drop to fill in the blank

✔ **Correct Answers:**

| Blank Position | Correct Argument |  |
|----------------|------------------|--|
| Left of ~      | outcome          |  |
| Right of ~     | treat            |  |
| **After `      | `**              |  |
| data           | df               |  |

**Complete Code:**

```
import library
library(AER)
2SLS model
ivreg(
 formula = outcome ~ treat | instrument,
 data = df
)
```

**Why This Syntax Is Correct:**

The `ivreg()` function from the **AER** package uses a specific formula notation for instrumental variables:

| Formula Part | Meaning |
|--------------|---------|
|--------------|---------|

|                              |                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>outcome ~ treat</code> | Left side of <code>~</code> : dependent variable<br>Right side of <code>~</code> : endogenous treatment variable |
| <code>***</code>             | instrument***                                                                                                    |
| <code>data = df</code>       | The dataset containing all variables                                                                             |

How 2SLS Works Here:

| Stage        | Equation                                                                            | What Happens               |
|--------------|-------------------------------------------------------------------------------------|----------------------------|
| First Stage  | <code>treat = <math>\pi_0</math> + <math>\pi_1</math>(instrument)</code>            | Predict treatment from the |
| Second Stage | <code>outcome = <math>\beta_0</math> + <math>\beta_1</math>(treat_predicted)</code> | Regress outcome on predi   |

The coefficient  $\beta_1$  on `treat` gives you the **causal treatment effect** (LATE) identified by the instrument.

Quick Memory Aid:

```
outcome ~ treatment | instrument
```

Read it as: "outcome depends on treatment, where treatment is instrumented by..."

This is the standard notation for 2SLS in R's `ivreg()` function!

Which of the following is NOT displayed in the output of running the `bal.tab()` function to assess variable balance?

Estimate of the causal treatment effect.

Standardized mean differences.

Sample sizes of the treatment and control groups.

The variance ratio.

Fill in the blanks below to make the statement about the relationship between bandwidth, sample size, and treatment effect in regression discontinuity design true.

As the bandwidth gets wider, the sample size **increases**, which can produce **more** precise estimates of the local average treatment effect (LATE). As the bandwidth gets narrower, the sample size **decreases**, which can yield a **less** precise estimate of the LATE.

Click or drag and drop to fill in the blank

Put the five steps of a propensity score analysis in the correct order below.

Step 1:

Step 2:

Step 3:

Step 4:

Step 5:

Estimate treatment effect or update propensity score model

Check initial overlap and balance

Model propensity scores

Re-check overlap and balance

Perform propensity score weighting

Click or drag and drop to fill in the blank

Put the five steps of a propensity score analysis in the correct order below.

- Step 1:
- Step 2:
- Step 3:
- Step 4:
- Step 5:

Click or drag and drop to fill in the blank

Fill in the blanks to make the following statements true.

The  assumption states that as long as we have accounted for all , we would obtain the same outcomes if the groups swapped treatment statuses.

The potential outcomes framework assumption called  assumes that an individual's  status does not influence other individuals in the sample.

The  assumption states that, for the included covariates, all subgroups of individuals have some probability of getting either treatment assignment.

Click or drag and drop to fill in the blank



Fill in the blanks to make the following statements true.

The **conditional exchangeability** assumption states that as long as we have accounted for all **confounders**, we would obtain the same outcomes if the groups swapped treatment statuses.

The potential outcomes framework assumption called **SUTVA** assumes that an individual's **treatment** status does not influence other individuals in the sample.

The **overlap** assumption states that, for the included covariates, all subgroups of individuals have some probability of getting either treatment assignment.

**outcome**   **selection bias**

Click or drag and drop to fill in the blank

### Correct Answers:

#### First statement:

The **conditional exchangeability** assumption states that as long as we have accounted for all **confounders**, we would obtain the same outcomes if the groups swapped treatment statuses.

#### Second statement:

The potential outcomes framework assumption called **SUTVA** assumes that an individual's **treatment** status does not influence other individuals in the sample.

#### Third statement:

The **overlap** assumption states that, for the included covariates, all subgroups of individuals have some probability of getting either treatment assignment.

## Quick Summary of These Key Assumptions

| Assumption                                            | Meaning                                                                                                                       |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Conditional Exchangeability</b>                    | After controlling for confounders, treatment assignment is independent of potential outcomes. (Also called "no confounding")  |
| <b>SUTVA</b> (Stable Unit Treatment Value Assumption) | One person's treatment doesn't affect another person's (no interference/spillover).                                           |
| <b>Overlap</b> (Common Support)                       | Every subgroup defined by covariates has a positive probability of receiving either treatment ( $0 < P(\text{treat}=1) < 1$ ) |

## Why These Matter

- **Conditional Exchangeability** → Allows causal identification after adjustment.
- **SUTVA** → Ensures we can define clear potential outcomes.
- **Overlap** → Ensures we have comparable individuals in both groups (critical for matching, weighting, etc.).

Violations of these assumptions are among the most common threats to causal inference validity.

Suppose we are interested in getting a sense of the degree of overlap between the treatment and control groups for a particular variable. Which of the below functions will produce a plot to visually inspect distributions for a variable?

```
weightit()
```

```
bal.plot()
```

```
love.plot()
```

```
bal.tab()
```

Imagine that you are interested in studying the effect of listening to music on exercise. Individuals are randomly assigned to either listen to music during exercise or to not listen to music during exercise. Activity trackers are used to measure calories burned during exercise. Which of the following statements about the scenario above is FALSE?

There are two potential outcomes for each individual: number of calories burned while exercising had they listened to music and number of calories burned while exercising had they NOT listened to music.

Because individuals were randomly assigned to one of the two groups, we can only observe one of the potential outcomes.

The counterfactual outcome for an individual in the music group is the number of calories burned while exercising without music.

The observed outcome for an individual in the music group is the number of calories burned while exercising with music.

✓

## Correct Answer: Option 2

"Because individuals were randomly assigned to one of the two groups, we can only observe one of the potential outcomes."

---

### Why This Statement Is FALSE:

The reason we can only observe **one** potential outcome per individual is **not** due to random assignment—it's due to the **fundamental problem of causal inference**.

| What's Wrong                                 | Correct Understanding                                                                                                                                  |  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Attributes limitation to randomization       | We can only observe one outcome because a person <b>cannot simultaneously be in both groups</b> (with music AND without music)                         |  |
| Implies random assignment causes the problem | The fundamental problem exists <b>regardless of assignment mechanism</b> —even in observational studies, we still only see one outcome per person      |  |
| Confuses identification with observability   | Random assignment helps <b>identify</b> the average causal effect (makes groups comparable), but doesn't affect whether we can observe counterfactuals |  |

---

### Why the Other Statements Are TRUE:

| Option   | Explanation                                                                                                                                                                            |  |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Option 1 | ✓ <b>TRUE:</b> Every individual has two potential outcomes: $Y(1)$ = calories with music, $Y(0)$ = calories without music. This is the foundation of the potential outcomes framework. |  |
| Option 3 | ✓ <b>TRUE:</b> For someone in the music group, their <b>counterfactual</b> is what would have happened without music ( $Y(0)$ ).                                                       |  |
| Option 4 | ✓ <b>TRUE:</b> For someone in the music group, their <b>observed</b> outcome is calories with music ( $Y(1)$ ).                                                                        |  |

---

### Key Distinction:

| Concept | Description |
|---------|-------------|
|---------|-------------|

|                                         |                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Fundamental Problem of Causal Inference | We can never observe both $Y(1)$ and $Y(0)$ for the individual (logical impossibility)                          |
| Role of Random Assignment               | Ensures $E[Y(1) T=1] = E[Y(1) T=0]$ and $E[Y(0) T=1] = E[Y(0) T=0]$ , making causal effects <b>identifiable</b> |
| Observability                           | Limited by physics/logic, not by study design                                                                   |

Bottom Line:

The correct answer is **Option 2** because it incorrectly attributes the fundamental limitation (unobservable counterfactuals) to random assignment, when in reality this limitation exists **independently of how treatment was assigned**. Randomization solves the *selection bias* problem, not the *missing data* problem!

| Z | $y^0$ | $y^1$ | Y | $y^1 - y^0$ |
|---|-------|-------|---|-------------|
| 1 | 2     | 5     | 5 | 3           |
| 1 | 1     | 3     | 3 | 2           |
| 0 | 4     | 9     | 4 | 5           |
| 1 | 3     | 7     | 7 | 4           |
| 0 | 2     | 5     | 2 | 3           |

The average treatment effect (ATE) is one causal estimand we learned in this lesson. Given the table above, calculate two other causal estimands: average treatment effect on the treated (ATT) and average treatment effect on the control (ATC).

ATT = (3 + 2 + 4) / 3 = 3, ATC = (5 + 3) / 2 = 4

ATT = (5 + 3 + 7) / 3 = 5, ATC = (4 + 2) / 2 = 3

ATT = (2 + 1 + 3) / 3 = 2, ATC = (4 + 2) / 2 = 3

✓ Correct Answer: Option 1

ATT = (3 + 2 + 4) / 3 = 3, ATC = (5 + 3) / 2 = 4

How to Calculate:

The key is to use the **individual treatment effects** ( $y^1 - y^0$ ) and average them separately for treated and control groups.

ATT — Average Treatment Effect on the Treated

Rows where **Z = 1** (treated individuals): rows 1, 2, and 4

| Row | Z | $y^0$ | $y^1$ | $y^1 - y^0$ |
|-----|---|-------|-------|-------------|
| 1   | 1 | 2     | 5     | 3           |

|   |   |   |   |   |
|---|---|---|---|---|
| 2 | 1 | 1 | 3 | 2 |
| 4 | 1 | 3 | 7 | 4 |

**ATT** = (3 + 2 + 4) / 3 = 9 / 3 = **3**

**ATC — Average Treatment Effect on the Control**

Rows where **Z = 0** (control individuals): rows 3 and 5

| Row | Z | y <sup>0</sup> | y <sup>1</sup> | y <sup>1</sup> - y <sup>0</sup> |
|-----|---|----------------|----------------|---------------------------------|
| 3   | 0 | 4              | 9              | 5                               |
| 5   | 0 | 2              | 5              | 3                               |

**ATC** = (5 + 3) / 2 = 8 / 2 = **4**

**Why the Other Options Are Wrong:**

| Option   | Error                                                                                                                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option 2 | Uses the <b>observed outcome (y)</b> values instead of treatment effects (y <sup>1</sup> - y <sup>0</sup> ). ATT uses y <sup>1</sup> values (5, 3, 7) and A values (4, 2) — these are observed outcomes, not effects. |
| Option 3 | Uses the <b>counterfactual outcomes (y<sup>0</sup>)</b> for treatment instead of the treatment effects.                                                                                                               |

**Quick Summary:**

| Estimand | Formula                                     | Calculation   | Result     |  |
|----------|---------------------------------------------|---------------|------------|--|
| ATE      | Average of ALL individual treatment effects | (3+2+5+4+3)/5 | <b>3.4</b> |  |
| ATT      | Average of ITEs for <b>treated</b> (Z=1)    | (3+2+4)/3     | <b>3</b>   |  |
| ATC      | Average of ITEs for <b>control</b> (Z=0)    | (5+3)/2       | <b>4</b>   |  |

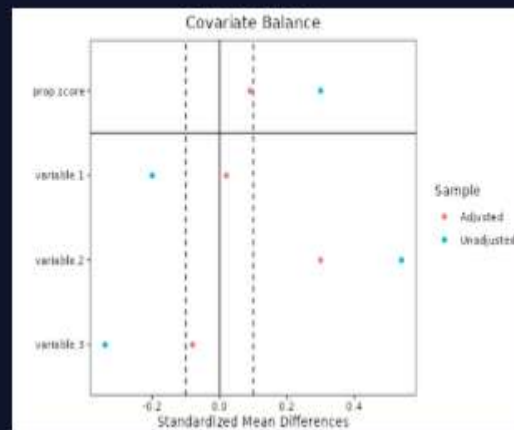
The crucial distinction: ATT and ATC are calculated from the **individual treatment effects column** (y<sup>1</sup> - y<sup>0</sup>), filtered by treatment status — not from raw observed outcomes!

The code below generates a Love plot to assess variable balance after performing propensity score weighting (IPTW). Based on the plot, which variables exhibit good balance after IPTW? Why?

```
import libraries
library(WeightIt)
library(cobalt)

perform IPTW in weightit object
wt_iptw <- weightit(
 formula = treatment ~ variable.1 + variable.2 + variable.3,
 data = weight_data,
 estimand = "ATE",
 method = "ps",
)

create love plot from weightit object
love.plot(x = wt_iptw, binary = "std", thresholds = c(m = 0.1))
```



None of the variables have good balance because the unadjusted SMD and adjusted SMD are not equal.

variable.1 and variable.3 both exhibit good balance because the adjusted SMD for both variables is within the range of -0.1 to 0.1.

Only variable.3 because the adjusted SMD is smaller than 0.1.

variable.1 and variable.2 because the unadjusted SMD for both variables is outside of the range of -0.1 to 0.1.

**Correct Answer:**

variable.1 and variable.3 both exhibit good balance because the adjusted SMD for both variables is within the range of -0.1 to 0.1.

## Explanation of the Love Plot

A **Love Plot** (from the cobalt package) is a visual tool to assess covariate balance **before** (unadjusted) and **after** (adjusted) weighting/matching.

- **X-axis:** Standardized Mean Difference (SMD).
- **Good balance** after IPTW: Adjusted SMDs (blue dots) should be **close to zero**, ideally **within  $\pm 0.1$** .
- Vertical dashed lines at  $\pm 0.1$  are common thresholds.

## Interpretation from the Plot:

- **variable.1** and **variable.3:** Their **adjusted** SMDs (blue points) fall within the **-0.1 to 0.1** range → **good balance** achieved after weighting.
- **variable.2:** Likely has an adjusted SMD outside this range → poor balance.

## Why the Other Options Are Incorrect

- "None of the variables..." → False, some variables improved.
- "Only variable.3..." → Too restrictive; variable.1 also looks good.
- Option about unadjusted SMDs → Irrelevant; we care about **adjusted** balance after IPTW.

## Code to Generate/Improve This

R

```
library(WeightIt)
library(cobalt)

IPTW
wt <- weightit(treatment ~ variable.1 + variable.2 + variable.3,
 data = weight_data, estimand = "ATE", method = "ps")

Love Plot
love.plot(wt, binary = "std", thresholds = c(m = 0.1))
```

**Rule of Thumb:** Aim for adjusted SMD < 0.1 and variance ratios close to 1.

Which of the following is NOT true about instrumental variable (IV) estimation?

Instrumental variable estimation produces accurate estimates of the treatment effect even when the instrument is weakly related to the treatment.

In IV estimation, treatment assignment and the treatment actually received are treated as separate factors.

Instrumental variable estimation helps account for bias even in the presence of unmeasured confounding variables.

In IV estimation, an instrument is a variable that is related to an outcome indirectly through another variable – typically the treatment variable.





# Correct Answer: Option 1

"Instrumental variable estimation produces accurate estimates of the treatment effect even when the instrument is weakly related to the treatment."

This statement is **NOT TRUE**.

## Why This Is FALSE:

Weak instruments are a **major problem** in IV estimation. Here's why:

| Problem with Weak Instruments                    | Consequence                                                                  |
|--------------------------------------------------|------------------------------------------------------------------------------|
| Low correlation between instrument and treatment | The first stage explains very little variation in the t                      |
| Biased estimates                                 | IV estimates become biased toward the OLS (bia                               |
| Poor inference                                   | Standard errors are unreliable; confidence interva coverage                  |
| Inconsistency                                    | Even with large samples, weak instruments don't produce consistent estimates |

**Rule of Thumb:** The F-statistic from the first stage should be **> 10** to avoid weak instrument problems. Below that, you risk severe bias.

## Why the Other Statements Are TRUE:

| Option   | Explanation                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option 2 | <b>✓ TRUE:</b> IV separates <b>treatment assignment</b> (instrument determines) from <b>treatment actually received</b> (endogenous variable). This is the core analysis. |
| Option 3 | <b>✓ TRUE:</b> One of IV's main strengths is addressing <b>unobserved confounding</b> . If the inst it can eliminate bias from unmeasured confounde                       |
| Option 4 | <b>✓ TRUE:</b> This correctly defines the <b>exclusion re</b> the instrument affects outcomes <b>indirectly</b> throug treatment, not directly.                           |

## Key IV Assumptions Recap:

| Assumption            | What It Requires                                               |
|-----------------------|----------------------------------------------------------------|
| Relevance             | Instrument must be <b>strongly correlated</b> with trea weak!) |
| Exclusion Restriction | Instrument affects outcome <b>ONLY</b> through treatm          |



|                        |                                               |
|------------------------|-----------------------------------------------|
| <b>Exchangeability</b> | No confounders between instrument and outcome |
| <b>Monotonicity</b>    | No "defiers" in the sample                    |

## Bottom Line:

The statement that is **NOT TRUE** is **Option 1**. Weak instruments lead to **inaccurate, biased, and unreliable** IV estimates. A strong first-stage relationship is essential for valid IV estimation!

Fill in the blanks to complete the statements below regarding what will happen if we apply inverse probability of treatment weighting (IPTW) for this sample.

| row | treatment group | propensity score |
|-----|-----------------|------------------|
| 1   | treatment       | 0.21             |
| 2   | treatment       | 0.85             |
| 3   | treatment       | 0.92             |
| 4   | control         | 0.15             |
| 5   | control         | 0.09             |
| 6   | control         | 0.79             |

IPTW will use the propensity scores in a formula to make treatment and control groups look . When the estimand is the average treatment effect (ATE), observations  will get  weights because they look different than those in their treatment assignment. Observations  will get  weights because they look similar to those in their treatment assignment.

1, 2, 3, and 6

1, 4, 5, and 6

less alike

1 and 6

higher

2, 3, 4, and 5

2 and 3

4 and 5

more alike

lower

Click or drag and drop to fill in the blank

### Correct Fill-ins:

IPTW will use the propensity scores in a formula to make treatment and control groups look **more alike**. When the estimand is the average treatment effect (ATE), observations **1 and 6** will get **higher** weights because they look different than those in their treatment assignment. Observations **2, 3, 4, and 5** will get **lower** weights because they look similar to those in their treatment assignment.

## Why This Is Correct

In **Inverse Probability of Treatment Weighting (IPTW)** for the **ATE**:

- The weight for each individual is typically: **Treated:**  $w = \frac{1}{PS}$  **Control:**  $w = \frac{1}{1 - PS}$
- **Rare events** (people who got treatment *against* their propensity score) receive **higher** weights.
- **Common events** (people whose treatment matches their propensity) receive **lower** weights (closer to 1).

## Application to the Table:

- **Row 1** (Treated, PS=0.21) → Unexpected → **High weight**
- **Row 6** (Control, PS=0.79) → Unexpected → **High weight**
- **Rows 2 & 3** (Treated, high PS) → Expected → **Low weight**
- **Rows 4 & 5** (Control, low PS) → Expected → **Low weight**

This up-weighting of "atypical" individuals helps balance the groups.

Fill in the blanks to complete the statements below regarding what will happen if we apply inverse probability of treatment weighting (IPTW) for this sample.

| row | treatment group | propensity score |
|-----|-----------------|------------------|
| 1   | treatment       | 0.21             |
| 2   | treatment       | 0.85             |
| 3   | treatment       | 0.92             |
| 4   | control         | 0.15             |
| 5   | control         | 0.09             |
| 6   | control         | 0.79             |

IPTW will use the propensity scores in a formula to make treatment and control groups look **more alike**. When the estimand is the average treatment effect (ATE), observations **1 and 6** will get **higher** weights because they look different than those in their treatment assignment. Observations **2, 3, 4, and 5** will get **lower** weights because they look similar to those in their treatment assignment.

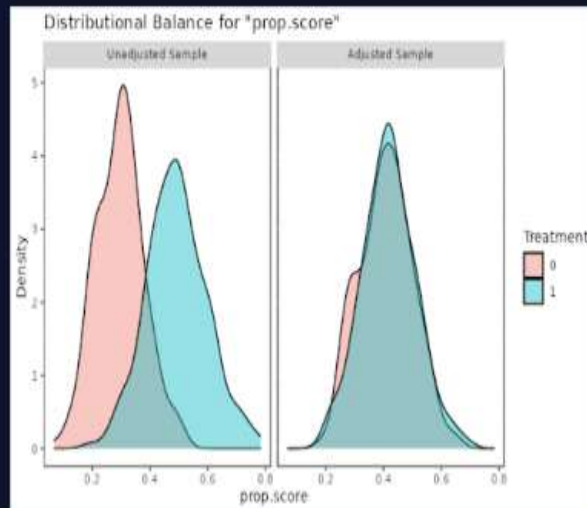
**1, 2, 3, and 6**   **1, 4, 5, and 6**   **less alike**   **2 and 3**   **4 and 5**

Click or drag and drop to fill in the blank

The following code was used to perform propensity score weighting using a dataset called `test_data`.

```
import libraries
library(WeightIt)
library(cobalt)
perform IPTW in weightit object
weight_obj <- weightit(
 formula = treatment ~ var1 + var2,
 data = test_data,
 estimand = "ATE",
 method = "ps"
)
```

Suppose that we want to generate the following plot to assess the balance of propensity scores before AND after propensity score weighting.



Fill in the right arguments to the `bal.plot()` function below that would produce the plot above.

`bal.plot(x =` `, which =` `, var.name =` `)`

`"var1"`

`"prop.score"`

`weight_obj`

`"adjusted"`

`test_data`

`"both"`

Click or drag and drop to fill in the blank

**Correct Code:**

R

```
bal.plot(x = weight_obj,
 which = "both",
```

```
var.name = "prop.score")
```

---

## Explanation of Arguments

- **x = weight\_obj**: This is the object created by `weightit()` — it contains the propensity scores and weights.
- **which = "both"**: Tells `bal.plot()` to show **both** the unadjusted (original) and adjusted (weighted) distributions side-by-side.
- **var.name = "prop.score"**: Specifies that we want to plot the **propensity score** variable.

---

## Why This Produces the Desired Plot

The resulting plot is a **density plot** comparing the distribution of propensity scores between treated and control groups:

- **Left panel**: Unadjusted sample (before weighting)
- **Right panel**: Adjusted sample (after IPTW weighting)

Good balance is indicated when the two distributions (treatment vs control) become much more similar in the **adjusted** panel.

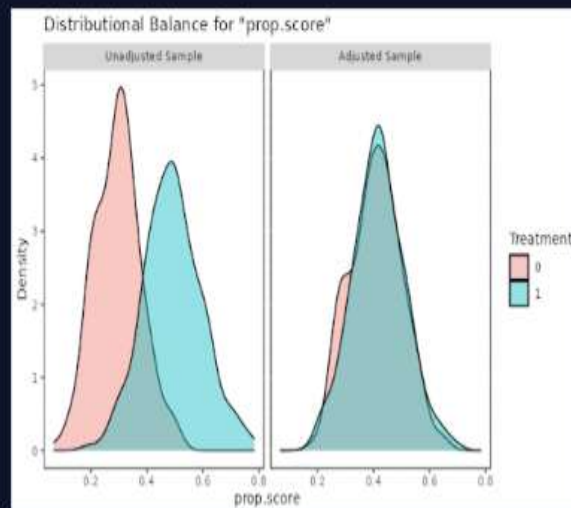
---

## Full Context Code

The following code was used to perform propensity score weighting using a dataset called `test_data`.

```
import libraries
library(weightit)
library(cobalt)
perform IPTW in weightit object
weight_obj <- weightit(
 formula = treatment ~ var1 + var2,
 data = test_data,
 estimand = "ATE",
 method = "ps"
)
```

Suppose that we want to generate the following plot to assess the balance of propensity scores before AND after propensity score weighting.



Fill in the right arguments to the `bal.plot()` function below that would produce the plot above.

`bal.plot(x =`  `weight_obj` `, which =`  `"both"` `, var.name =`  `"prop.score"` `)`

`"var1"`

`"adjusted"`

`test_data`

Click or drag and drop to fill in the blank

R

Which of the following does the Difference in Differences model estimate?

Average treatment effect on the treated (ATT)

Effect of time alone

Average treatment effect on the control (ATC)

✓

Correct Answer: Average treatment effect on the treated (ATT)

Why This Is Correct:

The Difference-in-Differences (DiD) model estimates the **Average Treatment Effect on the Treated (ATT)**, which is the causal effect of the treatment specifically for those who received it.

| Concept           | Explanation                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------|
| What DiD Does     | Compares the <i>change</i> in outcomes for the treated the <i>change</i> for the control group |
| Who It Applies To | Individuals who actually received the treatment                                                |
| Formula           | $DiD = (Y_{T,post} - Y_{T,pre}) - (Y_{C,post} - Y_{C,pre}) = ATT$                              |

How DiD Works:

|               | Pre-Treatme<br>nt | Post-Treatme<br>nt | Change                      |
|---------------|-------------------|--------------------|-----------------------------|
| Treated Group | $E[Y_0]$          | $E[Y_1]$           | $\Delta T$                  |
| Control Group | $E[Y_0]$          | $E[Y_0]$           | $\Delta C$                  |
| DiD Estimate  | —                 | —                  | $\Delta T - \Delta C = ATT$ |

The key assumption is **parallel trends**: in the absence of treatment, the treated and control groups would have experienced the same change over time.

Why the Other Options Are Wrong:

| Option                                          | Problem                                                                                                                       |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| "Effect of time alone"                          | The time effect is captured by the <code>time</code> coefficient regression — it's controlled for, not estimated as t effect. |
| "Average treatment effect on the control (ATC)" | DiD identifies effects for the treated group, not the We cannot estimate what would happen to the co they were treated.       |

Quick Summary:

| Estimand    | Estimated by DiD? | Reason                                                                         |  |
|-------------|-------------------|--------------------------------------------------------------------------------|--|
| ATT         | ✓ YES             | DiD tells us the effect of treatment for those who actually received it        |  |
| ATC         | ✗ NO              | Cannot infer effects for untreated units without strong additional assumptions |  |
| ATE         | ✗ NO (usually)    | Requires extrapolation beyond the treated population                           |  |
| Time effect | ✗ NO              | This is a control variable, not the target parameter                           |  |

Bottom Line:

The DiD estimator identifies the **Average Treatment Effect on the Treated (ATT)** under the parallel trends assumption. It tells us: *"How did the treatment affect those who actually received it, relative to what would have happened in their absence?"*

```
library(WeightIt)
library(cobalt)

weight_obj <- weightit(treatment ~ var1 + var2,
 data = test_data,
 estimand = "ATE",
 method = "ps")

Balance plot for propensity score
bal.plot(x = weight_obj,
 which = "both",
 var.name = "prop.score")
```



Assume that propensity score weighting/IPTW was performed on a dataset called `test_data` to reduce the imbalance of variables between the treatment and control groups. The IPTW weights are stored in the `iptw$weights` object. Both code blocks below produce estimates of the causal treatment effect from the IPTW-weighted regression model object `outcome_mod`, but the output will differ slightly.

```
import library
library(lmtest)
model causal treatment effect
outcome_mod <- glm(outcome ~ treatment + var1 + var2, data = test_data, weights = iptw$weights)
print results of model
coeftest(outcome_mod)
```

```
import libraries
library(lmtest)
library(sandwich)
model causal treatment effect
outcome_mod <- glm(outcome ~ treatment + var1 + var2, data = test_data, weights = iptw$weights)
print results of model
coeftest(outcome_mod, vcov. = vcovHC)
```

Which of the following describes how the second code block differs from the first code block?

- The second code block uses stabilized IPTW weights whereas the first block does not.
- The second code block will provide robust standard error estimates whereas the first will not.
- The second block models the outcome variable using logistic regression whereas the first block models the outcome using linear regression.

Correct Answer:

The second code block will provide robust standard error estimates whereas the first will not.

Detailed Comparison

| Aspect          | First Code Block               | Second Code Block      |
|-----------------|--------------------------------|------------------------|
| Model           | Same (glm() with IPTW weights) | Same                   |
| Standard Errors | Default (model-based)          | <b>Robust</b> (vcovHC) |



| Package Used | lmtest          | lmtest + sandwich                         |
|--------------|-----------------|-------------------------------------------|
| Purpose      | Basic inference | Better inference under heteroskedasticity |

## Why Use Robust Standard Errors?

In IPTW-weighted regressions:

- Weights introduce heteroskedasticity (unequal variance).
- Default standard errors can be biased or overly optimistic.
- **Robust (HC) standard errors** (from the sandwich package) are more reliable and are considered best practice.

The function `vcovHC()` computes **Heteroskedasticity-Consistent** covariance matrix.

## Recommended Final Code (Best Practice)

R

```
library(lmtest)
library(sandwich)

outcome_mod <- glm(outcome ~ treatment + var1 + var2,
 data = test_data,
 weights = ip

 tw$weights)

coeftest(outcome_mod, vcov = vcovHC(outcome_mod, type = "HC1"))
```

**Summary:** The two blocks fit the **same model**. The only meaningful difference is that the second block uses **robust standard errors** for more trustworthy inference.



